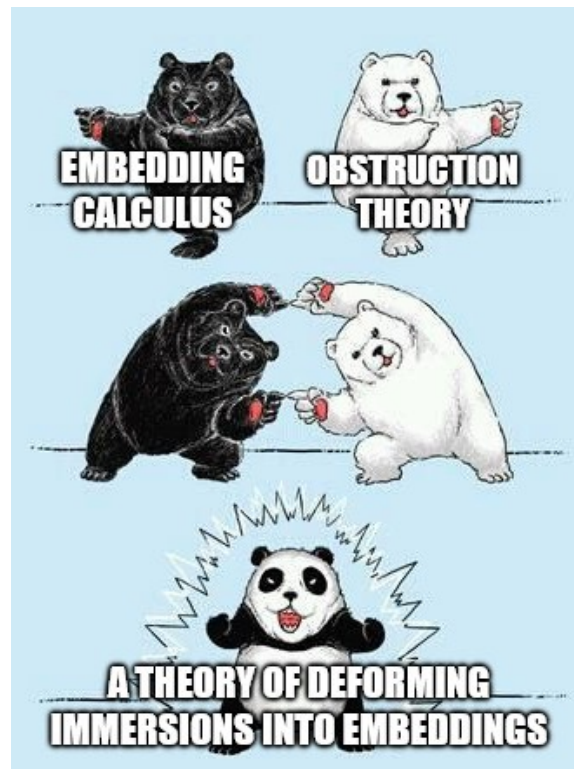


Embedding calculus, obstruction theory and immersions

MSc thesis



R.A. Vlierhuis

Supervised by Inbar Klang



Department of Mathematics
Vrije Universiteit Amsterdam
August 2026

EMBEDDING CALCULUS, OBSTRUCTION THEORY AND IMMERSIONS

R.A. VLIERHUIS

ABSTRACT. In this thesis, we will study the problem of when an immersion between two smooth manifolds M^m, N^n can be deformed (through immersions) into an embedding. The classical theory of Wall [Wal66] gives us some idea how to do this when $n = 2m$. We will discuss an alternative (and, to my knowledge, new) approach based on obstruction theory and embedding calculus that generalizes Wall's invariant and use our approach to recover a classical result of Smale [Sma59] on embeddings $S^m \rightarrow \mathbb{R}^{2m}$. We will spend much time setting up the obstruction theory from scratch, and we will discuss a lot of the classical theory that underlies the ideas of this thesis. At the end, suggestions are made for further research and for ways to use this new approach to obtain possible original results.

1. INTRODUCTION

Let M^m, N^n be smooth manifolds with $m < n$. One of the most important problems in geometric and differential topology is to determine the space $\text{Emb}(M, N)$ of *embeddings* from M to N . This is, in general, extraordinarily hard. However, with time, many tools have been developed to study this embedding space, at least in some specific cases, and we will see a number of them in action in this thesis. The guiding principle on our journey will be the following: instead of the very hard space of embeddings, we can also study the space of *immersions* $\text{Imm}(M, N)$. It turns out that we have reasonable machinery to study this space with homotopy theory, thanks to the work of Stephen Smale and Morris Hirsch (see chapter 3). We can think of immersions as maps $M \rightarrow N$ that locally look like embeddings, but globally might intersect themselves. The important question to ask thus becomes: when can we remove the self-intersections? That is, when can we deform an immersion into an embedding?

Classically, this problem was studied when $n = 2m$ by C.T.C. Wall [Wal66]. He defined an invariant that takes an immersion with transversal self-intersections and essentially counts the self-intersections with a smart choice of sign to take into account that some self-intersections might cancel each other out. The Wall invariant is a *complete* invariant when $m \geq 3$, meaning that the invariant is zero if and only if the immersion can be deformed to an embedding. This is great, however, there is also a number of problems. Firstly, the process starts by making an immersion self-transverse and counting the double points, which is hard to do for an arbitrary immersion. Secondly, it is not very clear how to generalize Wall's invariant to other dimension pairs. Some attempts have been made, see [Gra13], but the theory quickly gets very complicated.

Because of these issues, we will take a different approach. Our method uses *embedding calculus*, which is a essentially a homotopy-theoretic tool that describes

the difference between immersions and embeddings. By studying the so-called layers of embedding calculus, we will be able to give a conceptual way to think about deforming immersions into embeddings. The algebraic framework we will use is the language of *obstruction theory*. This is a theory that allows us to translate between topology - do certain maps making a diagram commutative exist? - and algebra - does a certain cohomology class vanish? In particular, this thesis contains a self-contained introduction to ordinary obstruction theory and develops an equivariant obstruction theory based on the work of Dror Farjoun and Dwyer-Kan ([DK84a], [Far87]). We will use these things in the rest of the thesis to formulate things algebraically and to help us prove statements.

Structure of the thesis. In Chapter 2, I will discuss obstruction theory, as mentioned above. Chapter 3 contains classical theory. I will explain the crucial theorem of Smale and Hirsch that allows us to classify immersions algebraically and give some examples, including Smale's famous result on sphere eversion (Theorem 3.12). Then I will define Wall's invariant and explain why we need a different obstruction for our purposes, and finally I will discuss Haefliger's theory of embeddings in the metastable range, which will serve as motivation and a sanity check for embedding calculus. In Chapter 4, there are all the theorems about embedding calculus we will need, and Chapter 5 contains our main result (Theorem 5.2), a new proof of the classical theorem by Smale which states that all embeddings $S^m \rightarrow \mathbb{R}^{2m}$ are regularly homotopic to the standard embedding. Finally, in Chapter 6 we discuss the potential of this method and give suggestions for further research.

Terminology. In this thesis, various terminology will be used. Under this sub-header I have tried to collect words whose meaning might not be immediately clear to every reader, in the hope of avoiding unnecessary confusion.

This thesis is about smooth manifolds. Whenever I speak of an embedding or immersion between manifolds I mean a smooth immersion, and whenever I speak of an isotopy or regular homotopy I mean a path in the space of smooth embeddings or immersions. A map between topological spaces is always continuous.

If n is a natural number, We say a topological space X with basepoint x is n -connected if $\pi_k(X, x) = 0$ for all $k \leq n$ and all $x \in X$. In particular, this requires X to be nonempty. We can also define a (-1) -connected space to simply be a nonempty space. If $f : X \rightarrow Y$ is a map and z some point in the homotopy fiber of f , by the notations $\pi_k(X \rightarrow Y, f)$ and $\pi_k(f, z)$ I will mean the homotopy groups of the homotopy fiber, and I will call these 'the homotopy groups of the map f '. Note that this may differ from indexing conventions used for relative homotopy groups. We say a map f is n -connected when the map f is surjective on π_n and an isomorphism on π_k for $k < n$. By the long exact sequence for homotopy groups of a fibration, this is equivalent to saying the homotopy groups $\pi_k(f, z) = 0$ for $k \leq n - 1$. (This is an unfortunate case of conflicting terminology: a map is n -connected if and only if its homotopy fiber is $(n - 1)$ -connected.)

For terminology relating to G -spaces and -actions, please refer to Section 2.2.

While introducing the equivariant obstruction theory, I will assume that the reader is familiar with the basic objects in model categories, namely weak equivalences, fibrations and cofibrations. If you are not familiar, the most important things to know are 1) these concepts mirror their analogues in ordinary homotopy theory and 2) cofibrations are defined to have the left lifting property with respect

to trivial fibrations (i.e. fibrations that are also weak equivalences) and fibrations have the right lifting property with respect to trivial cofibrations. For instance this means that if $i : X \rightarrow Y$ is any cofibration, $p : Z \rightarrow W$ is any trivial fibration, and there is a commutative square

$$\begin{array}{ccc} X & \longrightarrow & Z \\ \downarrow & \nearrow \text{dotted} & \downarrow \\ Y & \longrightarrow & W \end{array}$$

then the dotted lift exists. We also call an object fibrant if the unique map to the point is a fibration and cofibrant if the unique map to the point is a cofibration.

Acknowledgments. I could not have written this thesis without the support and encouragement of my supervisor Inbar Klang, for which I am very grateful. I would also like to thank Renee Hoekzema for agreeing to be second reader for this thesis, and I would like to thank everyone else with whom I have had enlightening or otherwise enjoyable mathematical discussions during the writing process. In particular, I will mention Greg Arone (who pointed out to me a number of places I was wrong), Valentijn van der Land (who shared some of his embedding calculus wisdom with me) and Job Winterdijk (who organized the lovely SOS student seminar, in which I presented some of this work).

2. OBSTRUCTION THEORY

In this section I will give a self-contained introduction to obstruction theory, since this concerns results that are well-known but do not have a clear source in the literature. We will need to think about equivariant obstruction theory as well as the ordinary case. Therefore, this section is organized as follows, following the paper of Dwyer and Kan [DK84a]. First we discuss ordinary obstruction theory, where we shall give more details than Dwyer and Kan. Then we introduce some notation and terminology regarding equivariant stuff, and finally then we set up equivariant obstruction theory. In this section, whenever I use the word ‘space’, I mean a Kan complex, and by a ‘subspace’ I mean a Kan subcomplex. In applications, we will be thinking about actual topological spaces (in fact, CW-complexes) and we apply the singular complex functor to get a Kan complex so that we can use the theory of this section.

2.1. Ordinary case. The central question of obstruction theory is the following: given a subspace $A \subset X$ and a map $f : A \rightarrow Y$, when can we extend f to a map $X \rightarrow Y$, and in how many distinct ways can we do this?

$$\begin{array}{ccc} A & \longrightarrow & Y \\ f \downarrow & \nearrow ? & \\ X & & \end{array}$$

It turns out that these questions have fairly clear homotopy-theoretic answers. The important idea is to decompose the lifting problem into smaller lifting problems along the *Postnikov tower* of Y .

Definition 2.1. The Postnikov tower of a space Y is a sequence of spaces

$$\cdots \rightarrow Q_n Y \rightarrow Q_{n-1} Y \rightarrow \cdots \rightarrow Q_2 Y \rightarrow Q_1 Y$$

where

- (1) every map $Q_n Y \rightarrow Q_{n-1} Y$ is a fibration
- (2) $Y = \lim_k Q_k Y$
- (3) $\pi_i(Q_n Y) = 0$ for $i > n$
- (4) the natural map $Y = \lim_k Q_k Y \rightarrow Q_n Y$ induces an isomorphism $\pi_i Y \rightarrow \pi_i Q_n Y$ for all $i \leq n$.

A crucial remark is that it follows from this definition that the fiber of $Q_n Y \rightarrow Q_{n-1} Y$ is an Eilenberg-MacLane space $K(\pi_n Y, n)$.

There are multiple ways to construct Postnikov towers. The reader may be familiar with the classical construction obtained by adding cells to Y to kill all necessary homotopy groups. We will discuss a different construction that fits better with the simplicial viewpoint. Recall that the n -skeleton of a simplicial complex, $\text{sk}_n Y \subset Y$, was the smallest subcomplex that contains all the n -simplices of Y .

Definition 2.2. The n -coskeleton of Y , $\text{cosk}_n Y$, is the simplicial set which has as its k -simplices maps $\text{sk}_n \Delta^k \rightarrow Y$, where Δ^k is the standard k -simplex. The degeneracy and face maps are induced from those on $\text{sk}_n \Delta$.

There is a natural map $a_n : Y \rightarrow \text{cosk}_n Y$ given on k -simplices by viewing a k -simplex of Y as a map $\Delta^k \rightarrow Y$ and sending this to its restriction to $\text{sk}_n \Delta^k$.

Proposition 2.3. *This natural map a_n induces an isomorphism on homotopy groups π_k for $k < n$, and the homotopy groups $\pi_k \operatorname{cosk}_n Y$ for $k \geq n$ vanish.*

Proof. The map a_n is clearly a bijection on k -simplices if $k \leq n$, and thus it induces an isomorphism on homotopy groups π_k for $k < n$. For $k > n$ the vanishing of $\pi_k \operatorname{cosk}_n Y$ is clear, because a map $\operatorname{sk}_n \Delta^k \rightarrow Y$ that is constant on the boundary of Δ^k is constant. For $k = n$, we show that any map $\partial \Delta^{n+1} \rightarrow Y$ can be filled by an $(n+1)$ -simplex. But this is clear, because an $(n+1)$ -simplex is a map $\operatorname{sk}_n \Delta^{n+1} = \partial \Delta^{n+1} \rightarrow Y$. $\square$

It is also not difficult to show that the obvious map $\operatorname{cosk}_n Y \rightarrow \operatorname{cosk}_{n-1} Y$ given by restricting to $(n-1)$ -skeleta is a fibration and that $Y = \lim_n \operatorname{cosk}_n Y$, and as such $Q_n Y := \operatorname{cosk}_{n+1} Y$ is a model for the Postnikov tower of Y .

2.1.1. *The canonical cocycle.* To avoid working with local coefficients, let us assume that our space Y is simple, that is, it is path connected and the fundamental group is abelian and acts trivially on all higher homotopy groups. This allows us to fix once and for all a basepoint $y_0 \in Y$ and simply write $\pi_n(Y)$ for $n \geq 2$, where it is understood that any $\pi_n(Y, y)$ is canonically isomorphic to $\pi_n(Y, y_0)$.

Let $C_*(\operatorname{cosk}_n Y, \operatorname{cosk}_{n+1} Y)$ be the relative simplicial chain complex. We note that for any $n \geq 2$, there is a canonical map $C_{n+1}(\operatorname{cosk}_n Y, \operatorname{cosk}_{n+1} Y) \rightarrow \pi_n(Y)$. Noticing that $\operatorname{sk}_n \Delta^{n+1} = \partial \Delta^{n+1}$, it sends an $(n+1)$ -simplex $\operatorname{sk}_n \Delta^{n+1} \rightarrow Y$ to its homotopy class in $\pi_n(Y)$. An $(n+1)$ -simplex coming from $\operatorname{cosk}_{n+1}(Y)$ extends to a map $\Delta^{n+1} \rightarrow Y$ and thus gets sent to zero in $\pi_n(Y)$, so this map is well-defined. Moreover, this map is a cocycle, i.e. it kills boundaries of $(n+2)$ -simplices. It thus gives a canonical element $k^{n+1} \in H^{n+1}(Q_{n-1} Y, Q_n Y; \pi_n(Y))$, which is known as the *canonical cocycle*.

Now let us go back to the lifting problem. The first lift we have to construct is

$$\begin{array}{ccc} A & \longrightarrow & Q_1 Y \\ f \downarrow & \nearrow ? & \\ X, & & \end{array}$$

but constructing such a lift is equivalent to constructing a lift in the diagram

$$\begin{array}{ccc} \Pi A & \longrightarrow & \Pi Q_1 Y = \Pi Y \\ f_* \downarrow & \nearrow ? & \\ \Pi X & & \end{array}$$

on fundamental groupoids, and this is already an algebraic and tractable problem.

Existence of the other lifts is controlled by the first statement of ordinary obstruction theory.

Theorem 2.4 (Ordinary obstruction theory, first statement). *If $A \rightarrow X$ is a cofibration, there is a dotted map*

$$\begin{array}{ccc} A & \longrightarrow & Q_n Y \\ f \downarrow & \nearrow & \downarrow \\ X & \longrightarrow & Q_{n-1} Y \end{array}$$

that makes the top triangle commute and the lower triangle homotopy commute rel A if and only if the cohomology class $h^{n+1} \in H^{n+1}(X, A; \pi_n(Y))$ obtained by pulling back $k^{n+1} \in H^{n+1}(Q_{n-1}Y, Q_nY; \pi_n(Y))$ vanishes.

Remark. In what follows, I will refer to a dotted map with these properties as a *lift* of the diagram, without further comment.

Notice that this obstruction class satisfies an obvious naturality in (X, A) , namely, if we have a map $(X', A') \rightarrow (X, A)$, the induced map $H^{n+1}(X, A; \pi_n(Y)) \rightarrow H^{n+1}(X', A'; \pi_n(Y))$ sends the obstruction class for the triangle

$$\begin{array}{ccc} A & \longrightarrow & Y \\ \downarrow & \nearrow \text{dotted} & \\ X & & \end{array}$$

to the one for the composite triangle

$$\begin{array}{ccccc} A' & \longrightarrow & X & \longrightarrow & Y \\ \downarrow & & & \nearrow \text{dotted} & \\ X' & & & & \end{array}$$

Let us show Theorem 2.4. The proof starts with the following lemma.

Lemma 2.5. *The commutative diagram*

$$\begin{array}{ccc} Q_nY & \longrightarrow & * \\ \downarrow & & \downarrow \\ Q_{n-1}Y & \xrightarrow{c} & K(\pi_nY, n+1) \end{array}$$

representing the canonical cocycle is a homotopy pullback.

Proof. Let X be any simplicial set, and let us have a commutative diagram

$$\begin{array}{ccccc} X & & & & \\ & \searrow f & & \searrow & \\ & & Q_nY & \longrightarrow & * \\ & & \downarrow & & \downarrow \\ & & Q_{n-1}Y & \xrightarrow{c} & K(\pi_nY, n+1). \end{array}$$

We should construct a map $g : X \rightarrow Q_nY$ making the diagram homotopy commute. If $k < n+1$, the coskeleton restriction $Q_nY \rightarrow Q_{n-1}Y$ induces a bijection on k -simplices, and so we know what to map a k -simplex of X to. If $k = n+1$, we need to do some work. Let σ be an $(n+1)$ -simplex of X , and let τ be its image under f . Then we know $c(\tau) = 0 \in \pi_nY$. Note what $c : Q_{n-1}Y \rightarrow K(\pi_nY, n+1)$ does on $(n+1)$ -simplices: it sends a map $\mu : \text{sk}_n \Delta^{n+1} = \partial \Delta^{n+1} \rightarrow Y$ to the homotopy class $[\mu] \in \pi_nY$. That is, in this case $\tau : \partial \Delta^{n+1} \rightarrow Y$ is filled by a $\tilde{\tau} : \Delta^{n+1} \rightarrow Y$, which we fix. We now define $g(\sigma)$ to be the $(n+1)$ -simplex in Q_nY defined by $\tilde{\tau}$. To make the map g into a simplicial map, on k -simplices σ with $k > n+1$, we define $g(\sigma)$ to be an appropriate lift of $f(\sigma)$ to Q_nY : we fill $f(\sigma) : \text{sk}_n \Delta^k \rightarrow Y$ to

a map $f(\tilde{\sigma}) : \text{sk}_{n+1} \Delta^k \rightarrow Y$ by filling all the faces $\text{sk}_n \Delta^{n+1} \rightarrow Y$ with the lifts we fixed before.

This makes the bottom triangle homotopy commute and is a unique choice up to homotopy. Indeed, all of our choices are forced except the specific choice of lifts $\tilde{\tau}$ for every τ . Choosing different lifts does not change g up to homotopy, because every $(n+1)$ -sphere in $Q_n Y$ can be filled with an $(n+2)$ -simplex. $\square$

Proof. of Theorem 2.4. We can now consider the composed square

$$\begin{array}{ccc} A & \xrightarrow{\quad} & * \\ \downarrow & \nearrow & \downarrow \\ X & \xrightarrow{\quad} & K(\pi_n Y, n+1) \end{array}$$

which clearly admits a lift if and only if the bottom map is nulhomotopic rel A . But that is the same thing as the cohomology class h^{n+1} vanishing.

We then exploit the homotopy pullback square of Lemma 2.5 to get a map $X \rightarrow Q_n Y$ as follows:

$$\begin{array}{ccccc} X & & & & \\ & \searrow & & \nearrow & \\ & Q_n Y & \xrightarrow{\quad} & * & \\ & \downarrow & & \downarrow & \\ & Q_{n-1} Y & \xrightarrow{\quad} & K(\pi_n Y, n+1), & \end{array}$$

where we now know that the outside square commutes up to homotopy, so we get a dashed map $X \rightarrow Q_n Y$ that is unique up to homotopy. This makes both triangles in

$$\begin{array}{ccc} A & \xrightarrow{\quad} & Q_n Y \\ \downarrow & \nearrow & \downarrow \\ X & \xrightarrow{\quad} & Q_{n-1} Y \end{array}$$

homotopy commute; the bottom right triangle homotopy commutes by construction, and the top left triangle homotopy commutes by a unicity argument in the fiber square of Lemma 2.5. Now we can make the top left square commute strictly using the homotopy extension property of (X, A) . $\square$

We can not only study whether a lift exists but also how many there are.

Theorem 2.6 (Ordinary obstruction theory, second statement). *Let $n \geq 2$ and suppose that h^{n+1} vanishes, that is, there exists a lift*

$$\begin{array}{ccc} A & \xrightarrow{\quad} & Q_n Y \\ f \downarrow & \nearrow & \downarrow \\ X & \xrightarrow{\quad} & Q_{n-1} Y \end{array}$$

as before. Then $H^n(X, A; \pi_n Y)$ acts freely and transitively on the homotopy classes of lifts; in particular, there is a (non-canonical) bijection between homotopy classes of lifts and $H^n(X, A; \pi_n Y)$.

Proof. The action is defined from the fact that $Q_n Y \rightarrow Q_{n-1} Y$ is a fibration with fiber $K(\pi_n Y, n)$ and this space has an obvious multiplication. We can multiply a lift fiberwise by a map $X \rightarrow K(\pi_n Y, n)$ that is null on A and obtain another valid lift, which defines the action. Now consider the obstruction to creating a homotopy between two lifts, f_0 and f_1 :

$$\begin{array}{ccc} X \times \partial I \cup A \times I & \longrightarrow & Q_n Y \\ \downarrow & \nearrow & \downarrow \\ X \times I & \longrightarrow & Q_{n-1} Y \end{array}$$

where the top map is $f_0 \cup f_1$ on $x \times \partial I$ and the fixed map $A \rightarrow Q_n Y$ on A , and the bottom map is constantly the fixed map $X \rightarrow Q_{n-1} Y$. The obstruction to a lift in this diagram existing is by Theorem 2.4 an element of $H^{n+1}(X \times I, X \times \partial I \cup A \times I; \pi_n Y)$. Since $A \rightarrow X$ was a cofibration, one can identify this with $H^{n+1}(\Sigma X, \Sigma A, \pi_n Y)$, where Σ denotes reduced suspension. This is canonically isomorphic to $H^n(X, A; \pi_n Y)$ (the isomorphism is given by the boundary map in an appropriate Mayer-Vietoris sequence) and the action of $H^n(X, A; \pi_n Y)$ on this by multiplying f_0 with the map $X \rightarrow K(\pi_n Y, n)$ is exactly the group operation of $H^n(X, A; \pi_n Y)$ (as the group operation on cohomology coincides with the multiplication on Eilenberg-MacLane spaces). It is now clear that the action of $H^n(X, A; \pi_n Y)$ is free and transitive; fixing a choice of lift the second statement follows. $\square$

2.1.2. *Relative case.* There is a completely analogous theory for the relative case, i.e. the question of whether there exist lifts in a square

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ B & \longrightarrow & Y \end{array}$$

where $A \rightarrow B$ is a cofibration and $X \rightarrow Y$ is a fibration. For this, we replace our Postnikov tower from Definition 2.1 by a relative version.

Definition 2.7. Given a fibration $X \rightarrow Y$, its Postnikov tower is a sequence of spaces

$$\cdots \rightarrow Q_n \rightarrow Q_{n-1} \rightarrow \cdots \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0 \rightarrow Q_{-1} = Y$$

where

- (1) every map $Q_n \rightarrow Q_{n-1}$ is a fibration
- (2) $X = \lim_k Q_k$
- (3) the natural maps $Q_n \rightarrow Y$ induce isomorphisms $\pi_i(Q_n) \rightarrow \pi_i(Y)$ for all $i > n + 1$ and a monomorphism for $i = n + 1$
- (4) the natural maps $X \rightarrow Q_n$ induce isomorphisms $\pi_i(X) \rightarrow \pi_i(Q_n)$ for all $i < n + 1$ and an epimorphism for $i = n + 1$

This has as a consequence that the fiber of $Q_n \rightarrow Q_{n-1}$ is a $K(\pi_n(X \rightarrow Y), n)$.

Let us explain how to construct the relative Postnikov tower. The construction is a slight variation of the one for the absolute tower. For $n \geq 1$, let Q_n be the “ $(n + 1)$ -coskeleton of X over Y ”, i.e. the simplicial complex with k -simplices

commutative diagrams

$$\begin{array}{ccc} \mathrm{sk}_{n+1} \Delta^k & \longrightarrow & X \\ \downarrow & & \downarrow \\ \Delta^k & \longrightarrow & Y \end{array}$$

with face and degeneracy maps coming from $\mathrm{sk}_{n+1} \Delta$ and Δ . This comes with an obvious map to Y by restricting to the bottom map in the square and with an obvious map from X by taking a k -simplex, mapping it to Y and then taking the skeleton. The properties on homotopy groups are easily verified similarly to before.

For $n = 0$, we need a slightly different space. Consider instead the “restricted 1-coskeleton of X over Y ” with k -simplices

$$\begin{array}{ccc} \mathrm{sk}_1 \Delta^k & \longrightarrow & X \\ \downarrow & & \downarrow \\ \Delta^k & \longrightarrow & Y \end{array}$$

where *the top map is nulhomotopic*. This will have the desired properties. In particular, we note that $Q_0 \rightarrow Y$ is a fibration that induces isomorphisms on higher homotopy groups and therefore the lifting problem

$$\begin{array}{ccc} A & \longrightarrow & Q_0 \\ \downarrow & \nearrow & \downarrow \\ B & \longrightarrow & Y \end{array}$$

is equivalent to the lifting problem on fundamental groupoids. Then we can attack the rest of the resulting lifting problems in the exact same way as before to obtain

Theorem 2.8 (Obstruction theory, relative case). *For every $n \geq 1$, there is a canonical cocycle $k^{n+1} \in H^{n+1}(Q_{n-1}, Q_n; \pi_n(X \rightarrow Y))$ such that the diagram*

$$\begin{array}{ccc} A & \longrightarrow & Q_n \\ \downarrow & \nearrow & \downarrow \\ B & \longrightarrow & Q_{n-1} \end{array}$$

(where $A \rightarrow B$ is a cofibration) has a lift if and only if its pullback $h^{n+1} \in H^{n+1}(B, A; \pi_n(X \rightarrow Y))$ vanishes.

Moreover, $H^n(B, A; \pi_n(X \rightarrow Y))$ acts freely and transitively on the set of homotopy classes of lifts. In particular there is a (non-canonical) bijection between the set of homotopy classes of lifts and $H^n(B, A; \pi_n(X \rightarrow Y))$.

2.2. Equivariant case. We shall also be interested in obstruction theory for diagrams of spaces with a group action.

Definition 2.9. A G -space is a space X with an action of a group G . That is, it is a continuous map $G \times X \rightarrow X$.

Remark. We usually imagine the group G to be a finite group with the discrete topology, but all results in the following section hold for an arbitrary topological group G .

We usually denote the image of (g, x) under the map $G \times X \rightarrow X$ by $g \cdot x$ or simply gx . A map $f : X \rightarrow Y$ between two G -spaces is *equivariant* if $f(gx) = g \cdot f(x)$. Given a G -space and a subgroup $H \leq G$, we can define the *space of fixed points*, $X^H := \{x \in X : \forall h \in H, h \cdot x = x\}$ with the subspace topology, and the *space of orbits*, X_H , which is the set of equivalence classes under the relation generated by $x \sim h \cdot y$ for all $h \in H$ with the quotient topology. An action is called *free* if it has no fixed points, i.e. $X^G = \emptyset$, and *transitive* if for any two $x, y \in X$ there exists $g \in G$ such that $x = gy$, i.e. $X_G = *$. A free and transitive action sets up a non-canonical bijection between the sets X and G (see if you can figure out how.)

We will try to find out if an equivariant lift exists in a diagram of equivariant maps of G -spaces of the form

$$\begin{array}{ccc} A & \longrightarrow & Y \\ f \downarrow & \nearrow ? & \\ X & & \end{array}$$

Our treatment of this topic will be based on the work of Dror Farjoun and Dwyer-Kan. Dwyer and Kan have [DK84a] constructed a straightforward generalization of the ordinary obstruction theory just described to diagrams of D -spaces, and this is the first thing we will discuss. Let S denote the category of simplicial sets.

Definition 2.10 (*D-spaces*). Let G be a finite group and D be a small category. A *D-space* is a functor $D^{op} \rightarrow S$. The category of D -spaces will be denoted $D\text{-Spaces}$ or S^D .

This definition includes the concept of G -spaces if we view G as a category with one object, where the arrows correspond to elements of G and composition is given by the operation of G .

The important result behind Dwyer and Kan's construction is the observation that one can set up an appropriate model category structure on $D\text{-Spaces}$.

Theorem 2.11 ([DK84b], Theorem 2.2). *There is a (closed, simplicial) model structure on S^D where the weak equivalences and the fibrations are defined objectwise. That is, a natural transformation $X \Rightarrow Y$ is a weak equivalence if and only if $X(d) \rightarrow Y(d)$ is a weak equivalence in S for all $d \in \text{ob} D$, and similar for fibrations. We will refer to this model structure as the free model structure on S^D .*

Remark. The fibrant objects in this model structure are exactly those D -spaces that are Kan complexes on every $d \in \text{ob}(D)$.

Given this model category structure, we can define a suitable cohomology theory on D -spaces, which will be the domain where our obstruction theory lives. Just like the model structure, this will be done 'objectwise'. Fix a ring R . From a D -space X we can form the diagram of *local n -chains* given by $RX_n : D^{op} \rightarrow R\text{-Mod}$ that sends d to the n -chains $C_n(X(d); R)$, where C_* is any reasonable chains functor. With the boundary map coming from the chains this assembles into a chain complex in the (abelian) category of D -diagrams in $R\text{-Mod}$. We denote this chain complex simply by $C_*(X)$.

We will also need relative cohomology. For this, let $A \rightarrow X$ be a cofibration. One can show that this implies that $C_*(A)$ is a direct summand of $C_*(X)$, and therefore $C_n(X)/C_n(A)$ is free for all n . The strategy for this is to write down explicitly what cofibrations are in this model; they are essentially given by retracts of transfinite

compositions of pushout diagrams that represent attaching a cell times a so-called free D -orbit. In the case that $D = G$, this is essentially saying that a cofibration is a retract of a so-called *free* relative G -CW complex.

Definition 2.12. Recall that a G -space is called *free* if the action has no fixed points. We call a (relative or absolute) G -CW complex *free* if it is constructed out of only cells of the form $G/\langle e \rangle \times D^n$. Every free absolute G -CW complex is a free G -space.

Then you can see explicitly that this statement about the chains must be true. We denote the relative chain complex by $C_*(X, A)$.

Let us now choose a functor $M : D^{op} \rightarrow R\text{-Modules}$. This will be our *coefficient system*. We define cochains with coefficients in M , $C^*(X; M)$ to be the R -module $\text{hom}_D(C_*(X), M)$ with the obvious induced coboundary map and similarly for relative cochains. (Note that this is no longer a D -diagram of R -modules.) Our cohomology groups will simply be the homology of these cochains. We will denote these $H^*(X; M)$ or sometimes $H_D^*(X; M)$ to avoid confusion with ordinary cohomology.

The most common example of a coefficient system is given by letting Y be a simple D -space, that is, all $Y(d)$ are simple. Then there is an obvious coefficient system for every $n \geq 1$ that is given by the n th homotopy groups, $\pi_n Y : D^{op} \rightarrow \text{Ab}$ by sending d to $\pi_n(Y(d))$.

Warning. The simplicity assumption includes that the spaces are *nonempty*, which is necessary because I can't define homotopy groups without being able to pick a basepoint. This means you should be quite careful what you write, since practical examples often do include empty spaces.

These groups satisfy reasonable axioms for a cohomology theory: they depend functorially on X or (X, Y) and on the coefficient system, satisfy long exact sequences for pairs, and reduce to ordinary cohomology when D is the one-point category.

Now to define obstruction theory for D -diagrams, we first need to construct a Postnikov tower in this setting. Fortunately, we can simply do this objectwise, i.e. if we have Y a (fibrant, simple) D -space, we can define $Q_n(Y)(d) := Q_n(Y(d))$. (Isn't that a gorgeous definition?) Recall that fibrations were defined objectwise, so this construction yields us a tower of fibrations

$$\cdots \rightarrow Q_n Y \rightarrow Q_{n-1} Y \rightarrow \cdots \rightarrow Q_2 Y \rightarrow Q_1 Y$$

as in the ordinary case.

We can now define the canonical cocycle, just as in the ordinary case, and again it is essentially doing the same construction as before objectwise, that is, send a relative $(n+1)$ -chain in $C_{n+1}(X, A)(d) = C_{n+1}(X(d), A(d))$ to the homotopy class generated by its image in $\pi_n(Q_n Y(d) \rightarrow Q_{n-1} Y(d)) \cong \pi_n(Y(d)) \cong \pi_n Y(d)$.

Remark. Two remarks about this construction:

- This definition is only for simple Y . You can make a similar definition when Y is not simple, but you will need cohomology with local coefficients.
- We use that a cofibration $A \rightarrow X$ also induces cofibrations $A(d) \rightarrow X(d)$ for every $d \in \text{ob} D$. This is an easy argument using the lifting property definition of a cofibration and the fact that weak equivalences and fibrations are pointwise.

The setup of obstruction theory for D -diagrams was completely analogous to that for ordinary spaces. One therefore hopes that the resulting theorems are also the same, and this turns out to be the case. First, notice that we can again translate lifting the diagram

$$\begin{array}{ccc} A & \longrightarrow & Q_1 Y \\ f \downarrow & \nearrow ? & \\ X & & \end{array}$$

to lifting the diagram of fundamental groupoids, where the fundamental groupoid of a D -space is defined (how else?) objectwise. The main theorems of obstruction theory also still hold.

Theorem 2.13 ([DK84a]). *Let $A \rightarrow X$ be a cofibration of D -spaces and let Y be a simple fibrant D -space. For every $n \geq 2$, there is a canonical class $h^{n+1} \in H_D^{n+1}(X, A; \pi_n(Y))$ such that the diagram*

$$\begin{array}{ccc} A & \longrightarrow & Q_n Y \\ \downarrow & \nearrow & \downarrow \\ X & \longrightarrow & Q_{n-1} Y \end{array}$$

has a lift if and only if its pullback vanishes.

Moreover, $H_D^n(X, A; \pi_n(Y))$ acts freely and transitively on the set of homotopy classes of lifts. In particular there is a (non-canonical) bijection between the set of homotopy classes of lifts and $H_D^n(X, A; \pi_n(Y))$.

Proof. (Sketch.) The proofs can be copied from the previous subsection once you show that the diagram cohomology theory we just defined is represented by the Eilenberg-MacLane diagrams $K(\pi_n Y, k)$ that arise from the diagram $\pi_n Y$. One can prove this in much the same way as one usually proves the representability theorem; namely, by observing that maps into the Eilenberg-MacLane space form a cohomology theory and then concluding they must agree on nice spaces. Some interesting remarks about representability of cohomology theories for D -spaces are made in [Far87]. $\square$

There is one major caveat: the cofibration condition in the theorem above. Recall that the model structure on D -Spaces had weak equivalences and fibrations defined objectwise. It was briefly mentioned earlier that this means cofibrations are generated by attaching so called *free* D -orbits, which correspond to free G -cells if $D = G$. But this means that if your map $A \rightarrow X$ does not come from attaching only free cells, you cannot use the obstruction theory above. Another way to look at this is the following: if $D = G$, weak equivalences being objectwise means that a weak equivalence of G -spaces is just an underlying weak equivalence, and a fibration of G -spaces is an underlying fibration. Since we thus have more weak equivalences and fibrations than we would expect in a ‘reasonable’ model structure on D -Spaces, we will have fewer cofibrations. Therefore, a straightforward application of the theorem above to obtain an equivariant obstruction theory is not a great idea.

To fix this, we will need to make our G -spaces free. We will execute this with the following construction, the *diagram of fixed point sets*.

Definition 2.14. The *orbit category* $\mathcal{O}_G$ of G is the category whose objects are homogeneous spaces G/H (called *orbits*) for subgroups $H \leq G$, and whose morphisms are G -equivariant maps.

Given a G -space, we may associate to it an $\mathcal{O}_G$ -space in the following way.

Definition 2.15. If X is a G -space, the associated *diagram of fixed point sets* is $X^\mathcal{O} : \mathcal{O}_G^{op} \rightarrow S$ given by $X^\mathcal{O}(G/H) := X^H$, the H -fixed points of X . Equivalently, $X^\mathcal{O}(G/H) \cong \text{Hom}_{S^G}(G/H, X)$.

So a G -space determines a simplicial diagram on a larger category which encodes all the fixed-point data of G . A crucial result in equivariant homotopy theory is Elmendorf's theorem, which states that the homotopy theory of G -spaces is equivalent to the homotopy theory of $\mathcal{O}_G$ -diagrams. In particular, a map $f : X \rightarrow Y$ of G -spaces is a G -homotopy equivalence if and only if it induces ordinary homotopy equivalences on fixed point spaces.

Remark. For those who know model categories. To make this more precise, we can use the model structure on D -Spaces in which a morphism is a weak equivalence or a fibration if it is a weak equivalence or a fibration objectwise that we discussed previously. There is a model structure on G -Spaces where the weak equivalences and fibrations are weak equivalences or fibrations on all fixed point sets. If we select these model structures, Elmendorf's theorem can be stated as saying that the construction above fits into a Quillen equivalence $S^G \rightarrow S^{\mathcal{O}_G}$.

This tells us that replacing G -spaces by $\mathcal{O}_G$ -diagrams preserves everything we need to know. It is worth noting that we can also go back from an $\mathcal{O}_G$ -diagram X to a G -space by taking the space $X(G/\langle e \rangle)$ and noting that every $g \in G$ gives an equivariant self map $G/\langle e \rangle \rightarrow G/\langle e \rangle$, from which we recover the action on our G -space. The most important thing about this construction, however, is that it turns maps of G -spaces that are not cofibrations in the free model structure on S^G into cofibrations.

Proposition 2.16. *If we have a map $f : X \rightarrow Y$ of topological spaces with a G -action that is a retract of a relative G -CW pair, the induced map $f^\mathcal{O} : \text{Sing } X^\mathcal{O} \rightarrow \text{Sing } Y^\mathcal{O}$ is a cofibration.*

Proof. (Sketch.) The free cells that we can attach in the free model structure on $\mathcal{O}_G$ -Spaces to get cofibrations are exactly the orbits G/H that we can attach in order to get a relative G -CW complex. (In a sense, the category $\mathcal{O}_G$ is defined so that this is true.) $\square$

Remark. There is an analogous construction of a diagram of fixed point sets $\mathcal{O}_D$ when you start with any D -space instead of a G -space. See [Far87, Section 3].

Putting everything together, this tells us exactly what obstruction theory for G -spaces should be. Note that the condition that $A \rightarrow X$ be a retract of a relative G -CW complex in the previous proposition is quite reasonable, unlike the condition that we had before which required all cell attachments to be free. If we start with a lifting problem

$$\begin{array}{ccc} A & \longrightarrow & Y \\ \downarrow & \nearrow & \\ X & & \end{array}$$

where $A \rightarrow X$ is a map as in the proposition, we can first replace it by the equivalent lifting problem on diagrams of fixed point sets:

$$\begin{array}{ccc} A^\mathcal{O} & \longrightarrow & Y^\mathcal{O} \\ \downarrow & \nearrow \text{dotted} & \\ X^\mathcal{O} & & \end{array}$$

and then note that $A^\mathcal{O} \rightarrow X^\mathcal{O}$ is a cofibration in the free model structure on $\mathcal{O}_G$ -Spaces, so we can apply Theorem 2.13 to get obstruction classes $h^{n+1} \in H_{\mathcal{O}_G}^{n+1}(X^\mathcal{O}, A^\mathcal{O}; \pi_n Y^\mathcal{O})$, assuming that the diagram $Y^\mathcal{O}$ is simple. The cohomology theory for G -spaces that arises in this way (by looking at the fixed point data) is called *Bredon cohomology* and was introduced in [Bre67].

2.3. Unwinding the definitions. In the rest of the thesis, we will be applying equivariant obstruction theory to embedding calculus. It turns out that many of the diagrams we encounter will consist of exclusively or almost exclusively spaces with a free action. In this section I will briefly spell out what the implications of the previous paragraphs are for this case and how the theory simplifies.

Let us assume for the rest of this section that A and X are free G -spaces and that $A \rightarrow X$ is a retract of a relative free G -CW complex. In this case, $A \rightarrow X$ is actually a cofibration in the free model structure on G -Spaces, and thus we can apply Theorem 2.13 directly without passing to diagrams of fixed points.

Unwinding the construction of the cohomology theory used in Theorem 2.13 when $D = G$, we will get the following. The chain complex for a G -space X will simply be your chain complex of choice for ordinary homology, with the G -action on chains that comes from the action of G on X . It is clear that the boundary maps respect the action, since taking the boundary and then applying an element of G is the same thing as first applying an element of G and then taking the boundary. A coefficient system will be an R -module with a G -action satisfying some axioms, which will turn out to be equivalent to a module over the group ring $R[G]$. One takes the cochain complex by considering just group homomorphisms from chains to the coefficients that respect the G -action, and then computes homology of this complex to R -module valued cohomology (note that there is no longer any G -action here).

Remark. We observe that this cohomology theory is not so exotic. In fact, on free G -CW complexes, we can show using a straightforward argument that this is simply the homology of the quotient. For this we use cellular chains. Let X be a free G -CW complex, and $Y := X/G$ its quotient with obvious cell structure induced from X . If our coefficient system A has a trivial G -action, an equivariant cochain on X sends every cell that is part of the same free G -cell to the same value in A . But an equivariant cochain on X is thus the same thing as an A -valued cochain on Y . It is easily verified that this gives an isomorphism of chain complexes, and thus an isomorphism on homology. Studying G -spaces by looking at the cohomology of the orbit space is a common technique in equivariant homotopy theory, and this is known as *Borel cohomology*.

Let us say we have a lifting problem

$$\begin{array}{ccc} A & \longrightarrow & Q_n Y \\ \downarrow & \nearrow & \downarrow \\ X & \longrightarrow & Q_{n-1} Y. \end{array}$$

Again assuming Y is a simple space, we get a coefficient system $\pi_n Y$ by considering the G -action on this homotopy group that comes from the G -action on Y . The canonical cocycle is now of course defined by sending a chain $(\Delta^{n+1}, \partial\Delta^{n+1}) \rightarrow (X, A)$ to the homotopy class in $\pi_n(Q_n Y \rightarrow Q_{n-1} Y) \cong \pi_n Y$ it generates, and this assignment respects the G -action because all maps in the diagram are equivariant.

Notably, if X is obtained from A by adding a single free G -cell, the canonical cocycle is determined by its value on one of the disks making up the free G -cell. This reflects the idea that if you can lift an equivariant map to one disk, you can simply use the action to move it around to all other disks in the free G -cell and thus obtain the equivariant extension you want.

2.3.1. An example. To see this equivariant obstruction for free diagrams in action, we will discuss an example. Consider the $\mathbb{Z}/2$ -equivariant diagram

$$(2.1) \quad \begin{array}{ccc} S^1 & \longrightarrow & S^2 \\ \downarrow & \nearrow & \\ S^3 & & \end{array}$$

where the solid maps are inclusions of the equator and all actions on the spheres are the antipodal actions. Clearly, one can find a non-equivariant lift in this diagram, because both equator inclusions will be nulhomotopic. So the zero map $S^3 \rightarrow S^2$ will make this triangle homotopy commute, and then we can straighten this to an actual lift using the homotopy extension property (as in the proof of Theorem 2.4). On the other hand, I claim that there is no equivariant lift in (2.1), and to prove this we shall apply our obstruction theory. Notice that S^3 can be built from S^1 by attaching first a free $\mathbb{Z}/2$ -2-cell and then a free $\mathbb{Z}/2$ -3-cell.

We start by trying to find out what lifts are possible in the triangle

$$\begin{array}{ccc} S^1 & \longrightarrow & Q_1 S^2 \simeq D^3 \\ \downarrow & \nearrow g & \\ S^3 & & \end{array}$$

Let us do this on the level of simplicial sets. To map $g : S^3 = \partial\Delta^4 \rightarrow Q_1 S^2$ while fixing S^1 , we need to determine first what g will do on 2-simplices. There is two 2-simplices of S^3 that I will call σ_2^+ and σ_2^- , the ‘top’ and ‘bottom’ hemispheres, which are conjugate to each other under the $\mathbb{Z}/2$ -action and which have as boundary the 1-simplex that goes around the equator. We are required to map σ_2^+ to a 2-simplex $g(\sigma_2^+) =: \tau^+$ in S^2 that also has as its boundary the 1-simplex that goes around the equator, and then we are required to map σ_2^- to the antipode, τ^- . Notice that then gluing τ^+ and τ^- gives a non-null map $f : S^2 \rightarrow S^2$, see Fig. 1.

Now, similarly to before, there is also two 3-simplices of S^3 that represent the hemispheres, which I will call σ_3^+ and σ_3^- . Notice that to extend a map $S^2 \rightarrow Q_1 S^2$ to a map $g : S^3 \rightarrow Q_1 S^2$, we need to send σ_3^+ to a 3-simplex in $Q_1 S^2$, i.e. a map



FIGURE 1. Starting with an map $\Delta^2 \rightarrow S^2$ that sends the boundary to the equator and then making it into an equivariant map $S^2 \rightarrow S^2$ by gluing on the antipode always results in a map of odd degree, in particular a map that is not null.

$\text{sk}_2 S^3 \rightarrow S^2$. But this needs to be done compatibly with the boundary, and the boundary of σ_3^+ should agree with the non-null map $f : S^2 \rightarrow S^2$. So the only reasonable thing to do is to map σ_3^+ to the 3-simplex of $Q_1 S^2$ that is given by $f : \text{sk}_2 \Delta^3 = S^2 \rightarrow S^2$.

Thus, any lift g will have the property that it sends σ_3^+ to a 3-simplex f in $Q_1 S^2$ that represents a nonzero homotopy class in $\pi_2 S^2$. But this is exactly what the canonical cocycle sees: it sends a 3-simplex of S^3 to the homotopy class of the 3-simplex of $Q_1 S^2$ that g maps it to. Therefore, the class $h^3 \in H_{\mathbb{Z}/2}^3(S^3, S^1; \pi_2 S^2)$ is non-zero for any choices of lifts you make, and thus a lift in the diagram (2.1) cannot exist.

In the upcoming sections, we will discuss various applications of this obstruction theory setup. We will use ordinary obstruction theory in combination with the theorem of Smale and Hirsch (Theorem 3.3) to classify immersions, and then we will use equivariant obstruction theory in combination with embedding calculus to define obstructions to deforming immersions into embeddings.

3. CLASSICAL THEORY

In this section, we will take a look at various classical techniques and ideas surrounding the study of immersions and embeddings, as a preparation for the introduction of embedding calculus and to set the stage for our own ideas. Let us start with a general goal that will recur throughout.

Goal. Given (smooth) manifolds M^m , N^n , $m \leq n$, study the space of (smooth) embeddings $\text{Emb}(M, N)$. In particular, we are interested in the set of connected components $\pi_0 \text{Emb}(M, N)$.

Definition 3.1. A path in $\text{Emb}(M, N)$ from e_1 to e_2 is called an isotopy from e_1 to e_2 . That is, an isotopy is a homotopy through embeddings.

This problem turns out to be very hard. In a sense, this is because embeddings cannot be studied locally: gluing together two embeddings does not necessarily give you an embedding. As an example, one can glue together two embeddings of an open 1-disk into $\mathbb{R}^2$ into the figure-8, which is not an embedding. An alternative way to phrase this is that the *presheaf* $V \mapsto \text{Emb}(V, N)$ that sends an open set $V \subset M$ to its embeddings in $\text{Emb}(V, N)$ is not a *sheaf*.

3.1. The theorem of Smale and Hirsch. Something that you might, therefore, try to do is to study the sheafification of this presheaf. Thinking about this, one realizes that elements of the sheafification are maps $V \rightarrow N$ that are locally embeddings of disks into N . That is, these things are the maps we know as immersions. This suggests the following potentially easier goal.

Goal. Study the space of smooth immersions $\text{Imm}(M, N)$, in particular the set $\pi_0 \text{Imm}(M, N)$ of connected components.

Definition 3.2. A path in $\text{Imm}(M, N)$ from f_1 to f_2 is called a regular homotopy from f_1 to f_2 . That is, a regular homotopy is a homotopy through immersions.

Now this problem is in fact way more accessible, and we have quite a satisfying theorem that tells us the homotopy type of $\text{Imm}(M, N)$. To explain the theorem, recall that an immersion is by definition a smooth map whose derivative is everywhere injective. This means that to an immersion $f : M \looparrowright N$ we can associate a bundle map $df : TM \rightarrow TN$ which is f on underlying spaces and df_x on tangent spaces, and this map will be injective on fibers. Such maps are often called *bundle monomorphisms* or *formal immersions* and the space of such maps is denoted $\text{Mon}(TM, TN)$. This construction defines us a map $d : \text{Imm}(M, N) \rightarrow \text{Mon}(TM, TN)$. Smale and Hirsch proved that this map is an equivalence.

Theorem 3.3 ([Hir59], [Sma59]). *The map d defined above is an equivalence. In particular, every formal immersion is homotopic to one that came from an immersion.*

Remark (Homotopy sheafifications). This theorem has an interpretation in terms of sheafifications as well. We can regard formal immersions as the *homotopy* sheafification of embeddings. As with any homotopy limit, there is a canonical map from the strict sheafification to the homotopy sheafification, and this will be exactly d . A sketch of a proof that $\text{Mon}(T-, TN)$ is the homotopy sheafification of $\text{Emb}(-, N)$ is as follows: we can write the bundle monomorphism functor as sections of a suitable fiber bundle. This implies that it is a homotopy sheaf. On open balls,

$\text{Mon}(T\mathbb{R}^m \rightarrow TN) \simeq \text{Emb}(\mathbb{R}^m, N)$ because both are contractible, and this suffices to prove that $\text{Mon}(T-, TN)$ must be the homotopy sheafification of $\text{Emb}(-, N)$. The theorem above thus says that the homotopy sheafification of $\text{Emb}(-, N)$ is equivalent to the strict sheafification, or equivalently that $\text{Imm}(-, N)$ is a homotopy sheaf. There is an approach to embedding calculus using homotopy sheaves; this is covered in [dBW13].

The reason this is such a big deal is that the space $\text{Mon}(TM, TN)$ is quite reasonable to attack using standard techniques of homotopy theory. Before we explore how to do this, however, it is time to discuss some relevant spaces: Grassmannians and Stiefel manifolds.

3.2. Stiefel manifolds and Grassmannians. Let us simply start with the definitions. Let $m \leq n$ be integers. The facts in this section can all be found in any course on fiber bundles, for instance the notes used for the Mastermath course Algebraic Topology II [HR25].

Definition 3.4. The *Stiefel manifold* $V_m\mathbb{R}^n$ is the collection of m -frames of $\mathbb{R}^n$ (i.e. sets of m linearly independent vectors), topologized as a subset of $(\mathbb{R}^n)^m$. It has an obvious action of $O(m)$ by matrix multiplication. The quotient $V_m\mathbb{R}^n/O(m)$ is the *Grassmannian* (of m -planes in $\mathbb{R}^n$) $\text{Gr}_m\mathbb{R}^n$.

We can also construct Stiefel manifolds and Grassmannians for an arbitrary vector bundle. As with any construction on vector bundles, these will be primarily applied to tangent bundles of manifolds.

Definition 3.5. Let $E \rightarrow X$ be a vector bundle of rank n . The Stiefel manifold V_mE is a fiber bundle over X defined as the bundle obtained by replacing the fibers of the frame bundle $\text{Fr}(E)$ by $V_m\mathbb{R}^n$, which has an $O(n)$ -action coming from the $O(n)$ -action on $\mathbb{R}^n$. The Grassmannian $\text{Gr}_m E$ is defined similarly.

In what follows, we will be very interested in computing homotopy groups of these spaces. For this, we will repeatedly apply the long exact sequence of homotopy groups to a number of fiber bundles. The first one arises from the fact that $\text{Gr}_m\mathbb{R}^n$ is the quotient of $V_m\mathbb{R}^n$ by an $O(m)$ -action; in fact, this defines a principal $O(m)$ -bundle

$$(3.1) \quad O(m) \longrightarrow V_m\mathbb{R}^n \longrightarrow \text{Gr}_m\mathbb{R}^n.$$

To compute homotopy groups of $V_m\mathbb{R}^n$ we will nearly always use the following fiber sequence.

Proposition 3.6. *Let $0 \leq \ell \leq m$. The projection p to the first ℓ vectors is a fiber bundle*

$$V_{m-\ell}(\mathbb{R}^{n-\ell}) \rightarrow V_m(\mathbb{R}^n) \xrightarrow{p} V_\ell(\mathbb{R}^n).$$

Proof. This is a straightforward exercise. □

Noting that $V_k\mathbb{R}^k = O(k)$, we get as an immediate consequence (by choosing $m = n$) that

Corollary 3.7. *There is a fiber bundle*

$$O(n - \ell) \rightarrow O(n) \rightarrow V_\ell\mathbb{R}^n.$$

Remark. In fact, more is true. The group $O(n-\ell)$ acts freely on $O(n)$ by permuting the last $n-\ell$ coordinates, and $V_\ell \mathbb{R}^n$ is the quotient of this action. Furthermore, the space $O(k)$ is a double cover of $SO(k)$ for all k . It is not hard to also prove that $V_\ell \mathbb{R}^n$ is the quotient $SO(n)/SO(n-\ell)$.

Noting that $V_1 \mathbb{R}^k = S^{k-1}$ we get as an immediate consequence of Proposition 3.6 (by choosing $\ell = 1$) that

Corollary 3.8. *There is a fiber bundle*

$$V_{m-1} \mathbb{R}^{n-1} \rightarrow V_m \mathbb{R}^n \rightarrow S^{n-1}.$$

Since we have some knowledge about the homotopy groups of the spheres, this last corollary can already yield us results about homotopy groups of Stiefel manifolds.

Proposition 3.9. *The Stiefel manifold $V_m \mathbb{R}^n$ is $(n-m-1)$ -connected.*

Proof. By induction on m . For $m = 0$ the resulting space is $V_0 \mathbb{R}^n$, which is a point, which is (-1) -connected (i.e. nonempty). Suppose the result is true for $m-1$ and consider $V_m \mathbb{R}^n$. By the previous corollary, there is a long exact sequence of homotopy groups

$$\pi_{k+1} S^{n-1} \rightarrow \pi_k V_{m-1} \mathbb{R}^{n-1} \rightarrow \pi_k V_m \mathbb{R}^n \rightarrow \pi_k S^{n-1}.$$

Now since the sphere S^{n-1} is $(n-2)$ -connected, we get that $\pi_k V_{m-1} \mathbb{R}^{n-1} \rightarrow \pi_k V_m \mathbb{R}^n$ is an isomorphism for $k < n-2$ and surjective for $k = n-2$. By the induction hypothesis, $V_{m-1} \mathbb{R}^{n-1}$ was $(n-1-(m-1)-1) = (n-m-1)$ -connected, and since $m \geq 1$, we have that $n-m-1 \leq n-2$. Therefore, for every $k \leq n-m-1$ we have that

$$0 = \pi_k V_{m-1} \mathbb{R}^{n-1} \rightarrow \pi_k V_m \mathbb{R}^n$$

is at least surjective. This completes the proof. $\square$

This has the following notable consequence for the homotopy groups of the orthogonal groups.

Corollary 3.10. *The inclusion $O(m) \hookrightarrow O(n)$ is $(m-1)$ -connected.*

Proof. Combine the previous proposition with Corollary 3.7. $\square$

This shows that these homotopy groups stabilize as you go towards infinity. Therefore low homotopy groups of $O(n)$ agree with those of $\text{colim}_k O(k)$, usually simply denoted O . A classical theorem by Raoul Bott known as *Bott periodicity* computes the homotopy groups of O .

Now that we know the most important facts about Grassmannians and Stiefel manifolds, we can move on to attacking the space of bundle monomorphisms using homotopy theory.

Proposition 3.11. *The space $\text{Mon}(TM, TN)$ is naturally equivalent to the space*

$$\left\{ \begin{array}{ccc} & & \text{Gr}_m TN \\ & \nearrow & \downarrow \\ M & \xrightarrow{TM} & BO(m) \end{array} \right\}$$

of homotopy commutative diagrams with a choice of homotopy. Note that these are equivalent to commutative diagrams

$$\begin{array}{ccc} & E_\gamma := \mathrm{Gr}_m TN \times_{BO(m)} \tilde{P}BO(m) & \\ & \searrow \gamma & \downarrow \\ M & \xrightarrow{TM} & BO(m), \end{array}$$

where the map $\gamma : \mathrm{Gr}_m TN \rightarrow BO(m)$ is the classifying map of the tautological vector bundle over $\mathrm{Gr}_m TN$. The space $\tilde{P}BO(m)$ is the unbased path space of $BO(m)$ and the space E_γ is then defined as $\{(x, p) \in \mathrm{Gr}_m TN \times \tilde{P}BO(m) : \gamma(x) = p(0)\}$. The map $E_\gamma \rightarrow BO(m)$ sends $(x, p) \mapsto p(1)$.

Proof. Given a bundle monomorphism $(f, F) : TM \hookrightarrow TN$, there is an obvious associated map $c : M \rightarrow \mathrm{Gr}_m TN$ which takes $x \in M$ to $(f(x), F(T_x M))$, the image of the entire tangent plane at x . As F is fiberwise injective, this is an m -plane in TN . It is easy to see that pulling back the canonical vector bundle over $BO(m)$ to M via $\mathrm{Gr}_m TN$ along c will yield a rank m subbundle of f^*TN that is isomorphic to TM via the map (f, F) . Thus this choice of map yields a canonical homotopy commutative diagram as above with a choice of homotopy. Conversely, by taking pullbacks of the canonical bundle over $BO(m)$, we get that TM is isomorphic to a rank m subbundle of TN , and a choice of homotopy is exactly as choice of isomorphism, i.e. a choice of bundle mono $TM \hookrightarrow TN$. The second part of the proposition is because the vertical map is the fibration replacement of $\mathrm{Gr}_m TN \rightarrow BO(m)$. $\square$

In the following section, we will discuss some consequences of this Smale-Hirsch theory using this proposition and our obstruction-theoretic machinery.

3.3. Applications of the Smale-Hirsch theorem. To start, here is a great time to discuss the most notorious consequence of this theorem: Smale's result on eversions of spheres.

Theorem 3.12 (Sphere eversions). *All immersions $S^2 \rightarrow \mathbb{R}^3$ are regularly homotopic. In particular, the embedding with winding number 1 around the origin is regularly homotopic to the one with winding number -1 around the origin.*

It is not at all geometrically obvious how you would construct such a regular homotopy, so much so that when Smale discovered this result, his advisor, the aforementioned Raoul Bott, told him that it was obviously wrong. History would have it differently, and later much effort was put (successfully) into finding explicit models for these so-called sphere eversions. We shall not discuss this here, but let us have a look at the proof of the theorem. A lot of details on the explicit construction of these sphere eversions can be found in the article [FM80].

Proof. The crucial point is that $\pi_2 V_2 \mathbb{R}^3 = 0$, where $V_k \mathbb{R}^n$ is the space of k -frames in $\mathbb{R}^n$. This space can be described as $SO(n)/SO(n-k)$ according to the remark after Corollary 3.7. In this case we thus get $SO(3)/SO(1)$, but $SO(1)$ is just a point. So $\pi_2 V_2 \mathbb{R}^3 \cong \pi_2 SO(3)$ and it is easily checked that this is zero, for instance by noting that $SO(3) \cong \mathbb{R}P^3$.

We use Proposition 3.11, where we note that the map $\mathrm{Gr}_m TN \rightarrow BO(m)$ has $V_m TN$ as its homotopy fiber. Applying the second statement of obstruction theory,

we see that the number of lifts is controlled by the number of lifts on fundamental groupoids and by the group $H^2(S^2; \pi_2 V_2 T\mathbb{R}^3)$, as all higher cohomology groups will clearly vanish. But $\Pi_1 S^2$ is clearly trivial, so we see that lifts in the diagram should correspond to classes in $H^2(S^2; \pi_2 V_2 T\mathbb{R}^3)$. But $V_2 T\mathbb{R}^3 \simeq V_2 \mathbb{R}^3$ and its second homotopy group is zero as computed above. $\square$

In fact, the method I just described in this proof goes way further than this. In much the same way, one can prove for instance that immersions $S^m \rightarrow \mathbb{R}^{2m}$ correspond to homotopy classes $\pi_m V_m \mathbb{R}^{2m}$. This motivates the following definition:

Definition 3.13. Let $f, g : S^m \looparrowright \mathbb{R}^{2m}$ be two immersions. With a regular homotopy, we can modify g so that it agrees with f on a closed disk D inside S^m . Then we can consider the *Gauss maps* $f_* : S^m \setminus D \rightarrow V_m \mathbb{R}^{2m}$ and $g_* : S^m \setminus D \rightarrow V_m \mathbb{R}^{2m}$ coming from fixing a trivialization of the tangent bundle of $S^m \setminus D$ and then considering the derivative map. These we can glue together into a map $S^m \rightarrow V_m \mathbb{R}^{2m}$. Then the homotopy class of this map is $\Omega(f, g)$, the *Smale invariant*.

Remark (On basepoints). Smale's paper formulates all of this in terms of based regular homotopies. Fortunately for us, as long as $m < n$, based regular homotopy classes of based immersions are naturally in bijection with unbased regular homotopy classes of immersions. One can see this in multiple ways; concretely, if you have any unbased immersion $f : S^m \looparrowright \mathbb{R}^n$, you can deform it into a based one by $f_t(x) := f(x) - tf(x_0)$, and if you have any unbased homotopy between immersions, you can similarly create a based one by $H'_t(x) := H_t(x) - H_t(x_0)$. More generally, we can use the fact that the evaluation $\text{Imm}(M, N) \rightarrow N$ sending $f \mapsto f(m_0)$ for a fixed $m_0 \in M$ is a Serre fibration ($m < n$); the fiber of this evaluation is the space of based immersions $\text{Imm}_*(M, N)$ and when $N \simeq *$ (as it is here) this actually shows $\text{Imm}_*(M, N) \hookrightarrow \text{Imm}(M, N)$ is a weak equivalence.

By quite similar techniques, we can also show that unbased homotopy classes $S^m \rightarrow V_m \mathbb{R}^n$ are in natural bijection with $\pi_m(V_m \mathbb{R}^n)$, only up to a choice of basepoint for $V_m \mathbb{R}^n$ if $m = 1, n = 2$. We will identify these homotopy classes of unbased maps with elements of these homotopy groups without further comment.

On the obstruction theory side of things we can see the basepoint appear in the square as in

$$\begin{array}{ccc} * & \longrightarrow & \text{Gr}_m TN \\ \downarrow & \nearrow & \downarrow \\ M & \xrightarrow{TM} & BO(m) \end{array}$$

and it is clear that the appearance of this point has no bearing on the relevant cohomology groups.

Proposition 3.14 (Smale 1957). *Let $m < n$. The Smale invariant is a complete invariant, i.e. $f, g : S^m \rightarrow \mathbb{R}^n$ are regularly homotopic if and only if $\Omega(f, g) = 0$. Moreover, for a fixed f and $\Omega_0 \in \pi_m V_m \mathbb{R}^{2m}$, there exists an immersion $g : S^m \rightarrow \mathbb{R}^n$ such that $\Omega(f, g) = \Omega_0$.*

Proof. (Sketch.) This is essentially a reformulation of the second statement of ordinary obstruction theory, where we note that the obstruction class to two lifts (i.e. f and g) being homotopic that lives in $H^m(S^m, \pi_m V_m \mathbb{R}^n)$ is essentially constructed to send the m -cell of S^m to the Smale invariant in $\pi_m V_m \mathbb{R}^n$. $\square$

3.4. Some homotopy group computations. I will pause here to remark that what we just achieved is remarkable: we showed that to compute immersions $S^m \rightarrow \mathbb{R}^n$, it suffices to compute homotopy groups of some Stiefel manifolds. Those computations are, given the correct input, quite accessible and quite algebraic in nature. That is, we have replaced a difficult differential topology problem by a much simpler algebraic topology one. In this section, I will illustrate this by computing the groups $\pi_m V_m \mathbb{R}^{2m}$ (and therefore the regular homotopy classes of immersions $S^m \rightarrow \mathbb{R}^{2m}$), which will be very interesting. However, let us start with a warmup and prove the following classical fact.

Proposition 3.15 (Smale 1957). *All immersions $S^m \rightarrow \mathbb{R}^{2m+k}$ for $k > 0$ are regularly homotopic.*

Proof. As we know, it suffices to prove that $\pi_m V_m \mathbb{R}^{2m+k} = 0$. Recall that $V_m \mathbb{R}^n$ can be written as a quotient

$$SO(n-m) \rightarrow SO(n) \rightarrow V_m \mathbb{R}^n.$$

We are thus interested in the inclusion $SO(m+k) \rightarrow SO(2m+k)$. However, we know (Corollary 3.10) that the inclusion $SO(n) \rightarrow SO(N)$, $n \leq N$, is $(n-1)$ -connected. Thus, this map $SO(m+k) \rightarrow SO(2m+k)$ is $(m+k-1)$ -connected, that is, at least m -connected. Thus, in the long exact sequence

$$\begin{aligned} \pi_m SO(m+k) \rightarrow \pi_m SO(2m+k) \rightarrow \pi_m V_m \mathbb{R}^{2m+k} \\ \rightarrow \pi_{m-1} SO(m+k) \rightarrow \pi_{m-1} SO(2m+k) \end{aligned}$$

the leftmost map is surjective and the rightmost map is an iso. We conclude from exactness that $\pi_m V_m \mathbb{R}^{2m+k}$ is zero. $\square$

It is time for the computation one dimension lower. This computation was known already to Stiefel; the proof below is adapted from [Pae58], which contains many more computations of homotopy groups of Stiefel manifolds.

Proposition 3.16. *The group $\pi_m V_m \mathbb{R}^{2m}$ is $\mathbb{Z}/2$ for m odd ≥ 3 and $\mathbb{Z}$ for all other $m \geq 1$.*

Proof. First we prove the following

Claim. For all $\ell \geq 2$, we have $\pi_k(V_\ell \mathbb{R}^{k+\ell}) \cong \pi_k(V_2 \mathbb{R}^{k+2})$.

Proof of claim. We use the fiber sequence

$$V_\ell \mathbb{R}^{k+\ell} \rightarrow V_{\ell+1} \mathbb{R}^{k+\ell+1} \rightarrow S^{k+\ell}.$$

On homotopy groups we get a long exact sequence

$$\pi_{k+1}(S^{k+\ell}) \rightarrow \pi_k(V_\ell \mathbb{R}^{k+\ell}) \rightarrow \pi_k(V_{\ell+1} \mathbb{R}^{k+\ell+1}) \rightarrow \pi_k(S^{k+\ell})$$

and as long as $\ell \geq 2$, the outside two terms vanish. The claim follows immediately.

Now our task is to compute $\pi_m V_m \mathbb{R}^{2m} \cong \pi_m V_2 \mathbb{R}^{m+2}$, $m \geq 2$. We will use that $V_2 \mathbb{R}^k$ is diffeomorphic to STS^{k-1} and look at the Gysin exact sequence (with $\mathbb{Z}$ coefficients)

$$H^0 S^{m+1} \rightarrow H^{m+1} S^{m+1} \rightarrow H^{m+1}(STS^{m+1}) \rightarrow H^1 S^{m+1}$$

where the first map is cupping by the Euler class of TS^{m+1} , the second map is induced from the projection $STS^{m+1} \rightarrow S^{m+1}$, and $H^1 S^{m+1}$ is of course zero. (An introduction to the Gysin sequence is in [HR25]). This is thus an exact sequence

$$\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow H^{m+1}(STS^{m+1}) \rightarrow 0$$

where the first map is given by multiplication by the Euler characteristic of S^{m+1} . If m is even, S^{m+1} has Euler characteristic zero, so $H^{m+1}(STS^{m+1}) \cong \mathbb{Z}$, while if m is odd, S^{m+1} has Euler characteristic one and $H^{m+1}(STS^{m+1}) \cong \mathbb{Z}/2$.

Finally, notice that STS^{m+1} is an orientable $(2m+1)$ -manifold and thus by Poincaré duality $H^{m+1}(STS^{m+1}) \cong H_m(STS^{m+1})$. Note that all homotopy groups of STS^{m+1} below m vanish, so by the Hurewicz theorem we have $\pi_m(STS^{m+1}) \cong H_m(STS^{m+1})$.

This finishes the proof for all $m \geq 2$. For $m = 1$, it is clear that $V_1\mathbb{R}^2$ is simply a copy of S^1 , so we are done in that case as well. $\square$

Remark. It is not hard to show using similar methods that $\pi_k V_m \mathbb{R}^{2m} = 0$ for $k < m$.

After this computation we have completely classified immersions $S^m \looparrowright \mathbb{R}^{2m}$ up to regular homotopy once we find a single choice of immersion. Of course, such a choice of immersion exists, since we have the standard embedding $S^m \rightarrow \mathbb{R}^{m+1}$. We can also see this algebraically from Proposition 3.11 since $H^{k+1}(S^m; \pi_k V_m \mathbb{R}^{2m})$ can clearly never be nonzero (if $k < m$ the coefficient group is zero, if $k \geq m$ there is no cohomology).

3.5. The Wall invariant. Let us go back to our goal from the beginning of the chapter.

Goal. Given (smooth) manifolds M^m, N^n , $m \leq n$, study the space of (smooth) embeddings $\text{Emb}(M, N)$. In particular, we are interested in the set of connected components $\pi_0 \text{Emb}(M, N)$, that is, the set of isotopy classes of embeddings $M \hookrightarrow N$.

This problem turned out to be way too difficult to attack directly, so in the previous sections we turned to studying immersions instead. In this section, we will go back to the world of embeddings and try to use our knowledge of immersions to say something about embeddings. Our strategy will be as follows.

Strategy. First classify immersions up to regular homotopy, and then determine which immersions are actually regularly homotopic to an embedding. This will yield a classification of embeddings up to regular homotopy.

The first part has now been covered, which means it is time for the second part of the strategy and the main topic of this thesis: when is an immersion regularly homotopic to an embedding? This problem was first studied in generality by C.T.C. Wall [Wal66, Section 3], who set up an invariant to study this problem when the target manifold is twice the dimension of the source and proved that it is complete starting in dimension 3. In this section, I will define his invariant and explain why, despite a complete obstruction to deforming immersions into embeddings already being known, it is still a worthwhile endeavor to define new ones.

Remark (Historical remark). This thesis takes the problem of classifying embeddings as its starting point. It is worth noting that this was not Wall's original motivation for studying this obstruction. His motivation came from surgery theory, where one is trying to remove nonzero homotopy classes from a space by performing an operation called surgery. This requires you to represent your homotopy class by an embedding of a sphere, where a priori you can only make it an immersion.

Let M^m be a simply connected closed manifold, N^{2m} be connected. An immersion $M \looparrowright N$ can be deformed to have transversal self-intersections. This is done

by considering the map $M \times M \setminus \Delta_M \rightarrow N \times N$ and making it transverse to the diagonal Δ_N .

Remark. It is not actually obvious how to do this from standard Thom transversality for smooth maps. The easiest way to show that this is possible is by using a *multijet transversality theorem*, see [Mat70, Theorem 3.3], where one notices that this gives what we want if we look at 2-fold 0-jets. One then also has to make an argument that we can do this through immersions, but fortunately immersions are generic inside smooth maps, so this works without issue.

Wall's invariant may be defined as follows. (I have taken this explanation of the construction of the Wall invariant from [Gra13].) Let $f : M \looparrowright N$. We fix a basepoint $m_0 \in M$ such that $f^{-1}(f(m_0)) = \{m_0\}$, i.e. a basepoint that is not a self-intersection of f , and take also the associated basepoint $n_0 = f(m_0) \in N$. The Wall invariant will take values in a quotient of the group ring $\mathbb{Z}[\pi]$ where $\pi = \pi_1(N, n_0)$. To define this quotient, consider the *orientation character* of N , $\omega : \pi \rightarrow \{\pm 1\}$ which sends a loop to 1 if fixing an orientation at n_0 and then going around the loop preserves the orientation and to -1 if it reverses the orientation. Note that if N is orientable, ω vanishes. We then define a $\mathbb{Z}/2$ -action on the group ring $\mathbb{Z}[\pi]$ by sending $\sigma \in \pi$ to $(-1)^m \omega(\sigma) \sigma^{-1}$. If $a \in \mathbb{Z}[\pi]$, we denote the image under this action by $\bar{a}$. The Wall invariant will now live in the quotient

$$H_0(\mathbb{Z}/2; \mathbb{Z}[\pi]) = \frac{\mathbb{Z}[\pi]}{\{a - \bar{a} : a \in \mathbb{Z}[\pi]\}}.$$

To define the invariant, note that the assumption that M is closed and the transversality of the self-intersections implies there is only finitely many of them. We denote by $\Sigma(f)$ the set of double points of f , that is, unordered pairs $[x, y]$ such that $f(x) = f(y)$. Let (x, y) be a choice of ordering for a double point. For the ordered pair (x, y) we define a loop $\sigma_{(x,y)} \in \pi$ and a sign $\varepsilon_{(x,y)} \in \{\pm 1\}$. To define the loop, choose paths γ_x and γ_y from m_0 to x and y to m_0 respectively that avoid other self-intersection points. (Note that we can do this because the dimension of M is at least 2, since there is no closed simply connected manifolds of dimension 1) Then define $\sigma_{(x,y)}$ to be the homotopy class of the loop in N based at n_0 obtained by concatenating $f\gamma_x$ and $f\gamma_y$. Notice that $\sigma_{(y,x)} = \sigma_{(x,y)}^{-1}$.

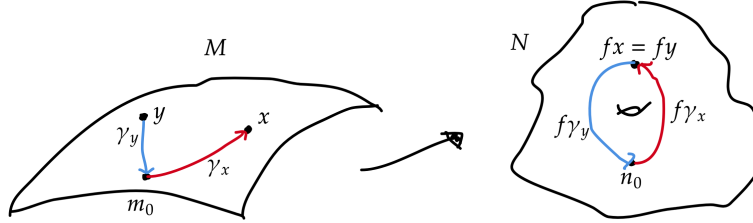


FIGURE 2. The element $\sigma_{(x,y)} \in \pi$ in the construction of the Wall invariant.

To define the sign, choose an orientation of M at m_0 and one of N at n_0 . The paths γ_x and γ_y then induce orientations on $T_x M$ and $T_y M$ by moving the chosen orientation at m_0 . The sign $\varepsilon_{(x,y)}$ equals 1 if the orientation of $df(T_x M)$ followed by $df(T_y M)$ agrees with the orientation on $T_{f(x)} N$ induced by moving the

chosen orientation at n_0 along $f\gamma_x$, and it equals -1 otherwise. (Recall that the self-intersection was transversal, so indeed $df(T_xM)$ and $df(T_yM)$ span $T_{f(x)}N$.) Observe that $\varepsilon_{(y,x)} = (-1)^m \omega(\sigma_{(x,y)}) \varepsilon_{(x,y)}$, where the factor $(-1)^m$ comes from swapping $df(T_xM)$ and $df(T_yM)$ and the factor $\omega(\sigma_{(x,y)})$ comes from transporting the orientation of N along $f\gamma_y$ instead of $f\gamma_x$.

Now the Wall invariant $\mu_W(f)$ is defined as the image in $H_0(\mathbb{Z}/2, \mathbb{Z}[\pi])$ of the formal sum

$$\sum_{[x,y] \in \Sigma(f)} \varepsilon_{(x,y)} \sigma_{(x,y)}$$

Note that the image in $H_0(\mathbb{Z}/2, \mathbb{Z}[\pi])$ does not depend on the choice of order of the x 's and y 's by construction and the assumption that M is simply connected means that this does not depend on the choice of γ_x and γ_y , so this is indeed well-defined.

Wall proves that

Theorem 3.17 ([Wal66], Theorem 3.1). *The Wall invariant $\mu_W(f)$ is a regular homotopy invariant. Furthermore, if $m \geq 3$, f is regularly homotopic to an embedding if and only if $\mu_W(f) = 0$.*

Note that it is easy to see that $\mu_W(f)$ vanishes when f is an embedding, since f will have no double points. The converse is proven using a variant of the Whitney trick [Whi44] for non-simply connected manifolds, which allows us to cancel self-intersections with ‘opposite orientation’. This is where the condition $n \geq 3$ comes from, since we cannot do the Whitney trick in dimension 4.

Example 3.18. The most classical case is when $M = S^m$ and $N = \mathbb{R}^{2m}$. In this case, the group π is equal to zero, and the $\mathbb{Z}/2$ -action on $\mathbb{Z}[\pi]$ reduces to the action on $\mathbb{Z}$ that sends $1 \mapsto (-1)^m$. That is, if m is odd, $\bar{a} = -a$ and thus the Wall invariant will live in $\mathbb{Z}/2$, and if m is even, $\bar{a} = a$ and the Wall invariant will be an integer.

Since the group π was zero, the only important part of the construction is the sign. The sign doesn’t matter if m is odd, in which case the Wall invariant is just the mod 2 self-intersection number. If m is even, the sign measures whether two self-intersection points have the same orientation or an opposite orientation (meaning you can cancel them out) and the resulting Wall invariant is the $\mathbb{Z}$ -valued intersection number.

At this point, one could consider the problem solved. We have found an invariant that completely determines whether an immersion $M^m \looparrowright N^{2m}$ is regularly homotopic to an embedding, and at first glance it seems readily computable. However, there are two major issues.

3.5.1. Computability. Recalling the construction of the Wall invariant, we note that the first step is to deform our immersion to have transversal self-intersections and then find the set of double points $\Sigma(f)$. The unfortunate part of this is that this is not such an easy thing to do explicitly. Perhaps if one can actually give a formula for the immersion, like if $M = S^m$ and $N = \mathbb{R}^{2m}$, it could be possible to do this explicitly, but it still would usually not be an easy task. Even worse, recalling the previous paragraphs, we will often construct our immersion classification from the abstract machinery of the Smale-Hirsch theorem and obstruction theory. There doesn’t appear to be a systematic way to obtain the number of self-intersections from this at all.

This presents a major obstacle to our strategy of classifying embeddings by starting with an immersion classification: we don't know how to compute which immersions are embeddings.

3.5.2. Generalizations. Another problem with this construction is that it is not so clear how to generalize it to cases where $n < 2m$. In this case, making your self-intersections transverse may also result in double circles and more generally double spheres. One could define a similar algebraic count of these double spheres and come up with a regular homotopy invariant of immersions $M^m \looparrowright N^{2m-k}$ that vanishes on embeddings. Some study of this is carried out by Grant [Gra13]. This quickly gets complicated, especially because the double spheres may form knots or links, and it's not immediately clear how to prove that your invariant is complete; at the very least, this will require a technical geometric argument.

Remark (Foreshadowing). In our framework, the reason that these cases are more complicated is because we might need to look at higher layers of embedding calculus (if $2n < 3(n+1)$) or at least consider multiple obstruction classes (in any case where $n < 2m$). See Chapter 4. We will see the bound $2n \geq 3(n+1)$ more often. It is called *Haefliger's metastable range* and will be discussed in the next section.

Both of these issues will be alleviated by looking at obstructions that come from embedding calculus and equivariant obstruction theory. Another advantage of our new methods will be that we might be able to say something about the number of embeddings in every regular homotopy class, i.e. a classification of embeddings up to isotopy. This is something that a Wall-style invariant will never be able to do.

Remark (The Hatcher-Quinn invariant). In [Gra13], the Wall invariant is compared with another obstruction to deforming immersions into embeddings, known as the Hatcher-Quinn invariant and introduced in [HQ74]. This invariant is better than the Wall invariant in the sense that it does generalize to at least some cases $n < 2m$, however, it suffers from the same computability issues because it requires a self-transverse immersion, and it is not clear how to generalize it to unmetastable dimension pairs. Klein-Williams give a new approach to the Hatcher-Quinn invariant in Part I of their papers on homotopical intersection theory [KW07] that does not rely on transversality. Their approach also uses cohomological obstructions to finding certain sections, so it is likely that it can be compared to our results, and this is an interesting topic for future research.

3.6. Haefliger's theory of embeddings. We will discuss one final piece of theory that will give the necessary context for embedding calculus and the lifting problem-style invariants that I will define. In this section, I will introduce some of the theory of Haefliger, which (at least phrased in modern language) is a more homotopy-theoretic and conceptual way to look at deforming immersions into embeddings.

Before we start, we need a definition.

Definition 3.19 (Isovariant & strictly isovariant maps). Let $f : X \rightarrow Y$ be an equivariant map of G -spaces. We say f is isovariant if for all subgroups $H \leq G$, we have $f^{-1}(Y^H) = X^H$. If X and Y are smooth manifolds on which G acts and f is smooth, we say f is strictly isovariant if, in addition, for all subgroups $H \leq G$ we have $df^{-1}(TY^H) = TX^H$.

If we pick the spaces $X = M \times M$ and $Y = N \times N$ with the $\mathbb{Z}/2$ -action given by flipping, the isovariance condition translates to $f^{-1}(\Delta_N) = \Delta_M$. The strict

isovariance condition translates to $df^{-1}(T\Delta_N) = T\Delta_M$. This is equivalent to saying that the restriction of df to the normal bundles of the diagonals is a bundle monomorphism. (To see this, view the normal bundle of Δ_N as the quotient $T(M \times M)|_{\Delta_N}/T\Delta_M$ and similarly for N .)

Let us denote the space of strictly isovariant maps by $\text{isoMap}_{\mathbb{Z}/2}(M \times M, N \times N)$. We then have the following theorem.

Theorem 3.20 ([Hae63], Théorème 1; reformulation as cartesian square in [GW99], Theorem 4.1). *There is a commutative square*

$$\begin{array}{ccc} \text{Emb}(M, N) & \longrightarrow & \text{isoMap}_{\mathbb{Z}/2}(M \times M, N \times N) \\ \downarrow & & \downarrow \\ \text{Map}(M, N) & \longrightarrow & \text{Map}_{\mathbb{Z}/2}(M \times M, N \times N) \end{array}$$

where the horizontal maps send $f \mapsto f \times f$ and the vertical maps are obvious inclusions.

If $2n \geq 3(m+1)$, this square is 0-cartesian, and if $2n > 3(m+1)$ it actually is 1-cartesian. The square is called n -cartesian if the comparison map from $\text{Emb}(M, N)$ to the homotopy pullback of the square is n -connected.

Remark (from [GW99]). Two points about this theorem:

- Haefliger originally proved this with just isovariant maps in the top right corner. Fortunately, Theorem 4.3 in [HH62] shows that the comparison map is 1-connected, so the result is still true in this formulation.
- It does not matter whether we choose $\text{Map}(M, N)$ to consist of all continuous maps $M \rightarrow N$ or only the smooth ones, since the inclusion is a weak homotopy equivalence thanks to the smooth approximation theorem.

This result somewhat indicates the strategy that we will follow in the rest of the thesis. The fact that this square is 0-cartesian means that to find out whether a map f is homotopic to an embedding, we look at the equivariant map $f \times f$ and see whether we can equivariantly homotope it into a strictly isovariant map. Whether we can do this gives a necessary and sufficient condition for deforming maps into embeddings.

Remark. The isovariance condition on a map $M \times M \rightarrow N \times N$ says that you send the diagonal to the diagonal, but nothing else. That is, you are forcing the map to be injective. The reason this method does not simply always work (but only in a range) is that requiring a lifting in this square is like making a bundle monomorphism injective on the underlying spaces, which is not quite the same as making the associated immersion injective.

The above result is the most well-known of Haefliger's paper, but there is also an analogue for deforming *immersions* into embeddings, instead of just any maps, and doing so with a regular homotopy instead of an ordinary homotopy. First we need to make a definition.

Definition 3.21 (Exit path space). Given a space Y , the space $\partial_2 Y$ is the space of paths in $Y \times Y$ that start at the diagonal and then leave and never return. In other words, $\partial_2 Y := \{p : I \rightarrow Y \times Y \mid p^{-1}(\Delta_Y) = \{0\}\}$. The topology is as a subspace of $\text{Map}(I, Y \times Y)$. This space has a natural $\mathbb{Z}/2$ -action that comes from flipping the coordinates of $Y \times Y$ and a natural (equivariant) map to $Y \times Y$ by evaluation at 1.

Remark. This definition will later be generalized to formulate Definition 4.3, hence the subscript 2.

Let us denote by X the space of pairs of equivariant dashed arrows that make the diagram

$$\begin{array}{ccc} \partial_2 M & \dashrightarrow & \partial_2 N \\ \downarrow & & \downarrow \\ M \times M & \dashrightarrow & N \times N \end{array}$$

homotopy commute. Note that $\text{Imm}(M, N)$ comes with a natural map to X ; composing a path in $\partial_2 M$ with $f \times f : M \times M \rightarrow N \times N$ might hit the diagonal again, but not arbitrarily close to zero by the fact that f is an immersion and the tubular neighborhood theorem. That is, we can construct the dashed map at least up to homotopy by contracting the path sufficiently. In much the same way, $\text{isoMap}(M \times M, N \times N)$ also comes with a natural map to X , by restricting the isovariant map to the normal bundle of the diagonal. We then have

Theorem 3.22 ([Hae63], Théorème 2, reformulation mine based on [GW99]). *There is a commutative square*

$$\begin{array}{ccc} \text{Emb}(M, N) & \longrightarrow & \text{isoMap}_{\mathbb{Z}/2}(M \times M, N \times N) \\ \downarrow & & \downarrow \\ \text{Imm}(M, N) & \longrightarrow & X \end{array}$$

where the maps are as discussed above. Again if $2n \geq 3(m+1)$, this square is 0-cartesian, and if $2n > 3(m+1)$ it actually is 1-cartesian.

Remark. Some remarks on this reformulation:

- The same comments as in the previous theorem apply, where we replaced Haefliger's original isovariant maps with these strictly isovariant ones while preserving the cartesian properties.
- Haefliger's original theorem mentions a homotopy H_t such that Δ_M is open inside $H_t^{-1}(\Delta_N)$ for all t . This condition is essentially there to avoid self-intersection points coming infinitesimally close to the diagonal. In the above formulation the condition has been replaced by requiring the equivariant square with the spaces $\partial_2 M$ and $\partial_2 N$ to commute, in order to obtain a nice formulation as a cartesian square.

Proof. (Sketch.) This version of the theorem follows from the previous theorem once you realize that the square

$$\begin{array}{ccc} \text{Imm}(M, N) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \text{Map}(M, N) & \longrightarrow & \text{Map}_{\mathbb{Z}/2}(M \times M, N \times N) \end{array}$$

is a homotopy pullback, essentially by the Smale-Hirsch theorem, Theorem 3.3. $\square$

Recall that at the beginning of this chapter, we talked about viewing immersions as an approximation to embeddings. Theorem 3.22 above suggests that the

homotopy pullback F of the square in the theorem is a *better* approximation, and we have maps

$$\begin{array}{ccc} & & F \\ & \nearrow & \downarrow \\ \mathrm{Emb}(M, N) & \longrightarrow & \mathrm{Mon}(TM, TN) \end{array}$$

One can wonder whether we can extend this upwards, that is, find spaces above F that approximate $\mathrm{Emb}(M, N)$ even better. This is essentially the idea of embedding calculus. A description of embedding calculus based on Haefliger squares is given in [\[GKW03\]](#).

4. EMBEDDING CALCULUS

Let me introduce the main tool that we shall use to study embeddings in this thesis: embedding calculus. The ideas in the following section are originally due to Goodwillie, Klein and Weiss [GW99], though I will sometimes use different formulations, following Boavida da Brito-Weiss [dBW13] and Krannich-Kupers [KK24].

4.1. Introduction. The fundamental idea of embedding calculus is that to study the space $\text{Emb}(M, N)$, it is a good idea to consider this as a particular value of a functor $\text{Emb}(-, N)$. In particular, if instead of all of M we plug some disjoint unions of open disks inside M into this functor, this can give us an approximation of the embedding space we are interested in. The hope is that the more disjoint disks we allow, the better the approximation gets. This hope is (under some condition) in fact a theorem and is referred to as *convergence* of embedding calculus. The name ‘embedding calculus’ refers to how this is a kind of polynomial approximation to the functor $\text{Emb}(-, N)$, i.e. a Taylor tower.

We start by stating the main theorem. The rest of the section will be devoted to motivating and interpreting the theorem and its direct consequences that we will use in this thesis. There will be no proofs as these are out of the scope of this work. Where appropriate I will add references to where one can read the proofs.

Theorem 4.1 ([GW99]). *There is a tower of spaces*

$$\begin{array}{ccc}
 & \cdots & \\
 & \downarrow & \\
 & T_k \text{Emb}(M, N) & \\
 & \downarrow & \\
 & T_{k-1} \text{Emb}(M, N) & \\
 & \downarrow & \\
 & \cdots & \\
 & \downarrow & \\
 \text{Emb}(M, N) & \xrightarrow{e_1} & T_1 \text{Emb}(M, N)
 \end{array}$$

$\nearrow^{e_k}$ $\nearrow^{e_{k-1}}$

where the functors $T_k \text{Emb}(-, N)$ are the k th Taylor approximations of $\text{Emb}(-, N)$, in the sense of Goodwillie. If the codimension $n - m \geq 3$, we have that the map e_k is $(k(n - m - 2) + 1 - m)$ -connected.

The implicit claim here is, of course, that these Taylor approximations are somehow easier to understand than the embedding functor itself. As it turns out, this is certainly the case. To be able to illustrate this in more detail, I will share one possible construction of the functors $T_k \text{Emb}(-, N)$ and thus the approximations to the embedding space.

Let $\mathbf{Mfd}$ be the ∞ -category of manifolds with morphisms embeddings. By the Yoneda lemma, we can consider $\text{Emb}(V, N)$ also as $\text{Map}_{\text{Psh}(\mathbf{Mfd})}(y_V, y_N)$, natural transformations of presheaves from $y_V := \text{Emb}(-, V)$ to $y_N := \text{Emb}(-, N)$. For every $k \geq 1$, we have full subcategories of $\mathbf{Mfd}$ called $\mathbf{Disk}^{\leq k}$ on all manifolds that are diffeomorphic to a disjoint union of k disks. (Note that the dimension of the disks is not relevant.) This gives us a natural restriction functor $\text{Psh}(\mathbf{Mfd}) \rightarrow$

$\text{Psh}(\mathbf{Disk}^{\leq k})$, and on mapping spaces this will induce our map

$$e_k : \text{Emb}(V, N) \rightarrow \text{Map}_{\text{Psh}(\mathbf{Disk}^{\leq k})}(y_V|_{\leq k}, y_N|_{\leq k}) =: T_k \text{Emb}(V, N).$$

Note that, by construction, the functor $V \mapsto T_k \text{Emb}(V, N)$ depends only on the values when V is a disjoint union of k disks (and the ways these values fit together). In other words, we say that this functor is *polynomial of degree k* , and the tower $\{T_k \text{Emb}(-, N)\}_k$ is a Taylor tower for the functor $\text{Emb}(-, N)$. One might now wonder what these functors $T_k \text{Emb}(-, N)$ actually are. For $k = 1$ this is easy to describe. Indeed, a natural transformation in $\text{Map}_{\text{Psh}(\mathbf{Disk}^{\leq k})}(y_V|_{\leq k}, y_N|_{\leq k})$ is basically a map $V \rightarrow N$ that is locally described by sending an embedding of a small disk inside V to an embedding of a small disk inside N . That is, this is a map $V \rightarrow N$ that is an immersion. One can similarly describe $T_2 \text{Emb}(V, N)$ as maps $V \rightarrow N$ that send embeddings of small pairs of disks to embeddings of small pairs of disks, that is, immersions that have no double points. The second layer of embedding calculus also corresponds to Haefliger's approximation to $\text{Emb}(M, N)$ described at the end of the previous chapter.

We have concluded that the first *layer* of embedding calculus is the space of immersions $\text{Imm}(M, N)$. On the other hand, if the conditions of Theorem 4.1 are satisfied, the map e_k will get more and more connected as k increases, since $n - m - 2 \geq 1$. Therefore we can conclude that the induced map $\text{Emb}(M, N) \rightarrow \text{holim}_k T_k \text{Emb}(M, N) =: T_\infty \text{Emb}(M, N)$ is a weak equivalence. We can thus say:

Slogan 4.2. Embedding calculus describes the difference between the space of immersions (which is well-understood) and the space of embeddings (which we are interested in).

4.2. The layers of embedding calculus. Of course, if we want to extract something useful out of the material in the previous section, we are going to need a better understanding of the spaces $T_k \text{Emb}(M, N)$. In particular, we need to know when you can lift an element of the $(k - 1)$ st layer to the k th layer. For this, we will make use of a layer description that I learned from the paper of Krannich and Kupers [KK24, Corollary 5.18], but which goes back to the work of Weiss [Wei99, Theorem 9.2, Example 10.3]. It tells us the difference between the layers in terms of a homotopy pullback square involving some spaces of equivariant diagrams.

Definition 4.3 (Spaces in the layer description). Let M a manifold, $k \geq 1$ an integer. The *fat diagonal* Δ_M^k is defined to be the subspace $\{(x_i)_i \in M^k : \exists i \neq j, x_i = x_j\} \subset M^k$, that is, the space of non-injective maps $\underline{k} \rightarrow M$. The *exit path space* $\partial_k M$ is the space of paths $p : I \rightarrow M^k$ that start inside Δ_M^k and never hit it again, that is, paths such that $p^{-1}(\Delta_M^k) = \{0\}$. The *ordered configuration space of k points in M* , denoted $F_k M$, is $M^k \setminus \Delta_M^k$, or also the space of injective maps $\underline{k} \rightarrow M$. More generally, the space of injective maps from a (finite) set S to M is denoted $F_S M$. Finally, we denote by $\partial_k^{\text{co}} M$ the space $\text{holim}_{S \subseteq \underline{k}} F_S M$.

All of these spaces inherit an obvious action of the symmetric group Σ_k by permuting the terms in the product M^k , and all natural maps between them are equivariant with respect to this action.

Definition 4.4 (More spaces in the layer description). We define the space X_k to be the space of pairs of equivariant dashed maps

$$\begin{array}{ccc} \partial_k M & \dashrightarrow & \partial_k N \\ \downarrow & & \downarrow \\ \partial_k^{co} M & \dashrightarrow & \partial_k^{co} N \end{array}$$

making the square homotopy commute. Compare the space X in Theorem 3.22. We define the space X'_k to be the space of triples of equivariant dotted maps

$$\begin{array}{ccc} \partial_k M & \dashrightarrow & \partial_k N \\ \downarrow & & \downarrow \\ F_k M & \dashrightarrow & F_k N \\ \downarrow & & \downarrow \\ \partial_k^{co} M & \dashrightarrow & \partial_k^{co} N \end{array}$$

making all squares homotopy commute. Note that there is a natural map $X'_k \rightarrow X_k$ by composing the two vertical arrows.

Theorem 4.5 ([KK24], Corollary 5.18 and section 5.10). *There is a homotopy pullback square*

$$\begin{array}{ccc} T_k \text{Emb}(M, N) & \longrightarrow & X'_k \\ \downarrow & & \downarrow \\ T_{k-1} \text{Emb}(M, N) & \longrightarrow & X_k. \end{array}$$

Compare Theorem 3.22 when $k = 1$.

Example 4.6. Let us unpack what happens in the most important case for this thesis, $k = 2$. The diagrams on the right hand side of the homotopy pullback then become

$$\begin{array}{ccc} \partial_2 M & \longrightarrow & \partial_2 N \\ \downarrow & & \downarrow \\ \partial_2^{co} M & \longrightarrow & \partial_2^{co} N \end{array}$$

and

$$\begin{array}{ccc} \partial_2 M & \longrightarrow & \partial_2 N \\ \downarrow & & \downarrow \\ F_2 M & \longrightarrow & F_2 N \\ \downarrow & & \downarrow \\ \partial_2^{co} M & \longrightarrow & \partial_2^{co} N. \end{array}$$

We note that $\partial_2^{co} M = \text{holim}_{S \subseteq \mathbb{Z}_2} F_S M = \text{holim}_{S \subseteq \mathbb{Z}_2} M = M \times M$. The fat diagonal in this case is simply the diagonal of M^2 , so $F_2 M = M^2 \setminus \Delta_M$, and $\partial_2 M$ can be modeled by the sphere bundle of the normal bundle of $\Delta_M \subset M^2$, which is equivalent to STM . It is now clear that the map $\partial_2 M \rightarrow \partial_2 N$ coming from an immersion $M \looparrowright N$ represents tangential data; in fact, the map $STM \rightarrow STN$

that appears in this way is simply the bundle monomorphism that you get as the derivative of f .

An equivariant lift to a diagram of the lower form then involves picking an equivariant map $F_2M \rightarrow F_2N$, i.e. a map $M \rightarrow N$ that sends distinct points in M to distinct points in N , i.e. an injective map, and the (homotopy) commutativity of the diagram ensures that our new map is homotopic to our original immersion in a way that respects the tangential data. In other words, finding a lift in this square can be interpreted as formally removing double points from your immersion.

As remarked in the previous chapter, Haefliger's theory is also about removing double points from maps or immersions; indeed, we saw that deforming maps or immersions to embeddings involved lifting to a space of strictly isovariant maps, which was like making a formal immersion injective. Indeed, these concepts turn out to be equivalent. This is illustrated for instance in [GKW03].

Since lifting from the first to the second layer corresponds to Haefliger's theory, we expect to recover the metastable range from Theorem 3.20 and Theorem 3.22. Indeed, we get this from Theorem 4.1: if we substitute in $k = 2$, we get that the map $e_2 : \text{Emb}(M, N) \rightarrow T_2 \text{Emb}(M, N)$ in this case is $2(n - m - 2) + 1 - m = (2n - 3m - 3)$ -connected. So whenever $2n \geq 3(m + 1)$, the map e_2 is at least 0-connected, and whenever $2n > 3(m + 1)$ the map e_2 is at least 1-connected. This recovers Theorem 3.22, at least when $m \geq 3$.

In particular, if $n = 2m$, $m \geq 3$ it is at least 0-connected, meaning surjective on connected components. If we unravel what this means, this is exactly saying that if you lift an immersion from T_1 to T_2 , it must be connected by a path in T_2 to an element of T_2 that came from an embedding. In particular, your immersion was regularly homotopic to an embedding.

A major advantage this setup has over that of Haefliger is that the spaces in this square are a lot more tractable than those in the Haefliger square. In fact, the original idea for the thesis was to generalize the Haefliger description with isovariant mapping spaces to the higher layers and to use isovariant homotopy theory to study these, but it turned out to be much easier to attack the equivariant squares we have here.

5. AN APPLICATION

In this section, we will give a proof of concept for studying embeddings using the methods described in this thesis. We shall find out which immersions $S^m \looparrowright \mathbb{R}^{2m}$, $m \geq 3$, are regularly homotopic to embeddings, recovering a result of Smale .

Recall from the previous section that to determine if an immersion $M^m \looparrowright N^{2m}$ is regularly homotopic to an embedding, we look for an equivariant lift in the associated diagram

$$(5.1) \quad \begin{array}{ccc} \partial_2 M & \longrightarrow & F_2 N \\ \downarrow & \nearrow & \downarrow \\ F_2 M & \longrightarrow & N \times N. \end{array}$$

We make the following important observation.

Proposition 5.1. *The inclusion $\partial_2 M \rightarrow F_2 M$ can be modeled as a free $\mathbb{Z}/2$ -CW complex.*

Proof. (Sketch.) Model this inclusion as an inclusion of $S\nu(\Delta_M)$ into $M \times M \setminus U$, where U is an open tubular neighborhood of Δ_M . We can find using standard techniques of differential topology a cell structure on $\Delta_M \cong M$ and on $M \setminus D^m$. Our cell structure on $F_2 M$ will be constructed by starting with a 0-simplex x of Δ_M and attaching to the sphere $S\nu_x \Delta_M$ the constructed cell structure on $M \setminus D^m$ along the identity on S^{m-1} and another copy of the constructed cell structure on $M \setminus D^m$ along the antipode. This will ensure that the action ‘swaps the coordinates’. We then repeat this procedure first for all 0-simplices, then multiplied by a 1-disk for 1-simplices, et cetera. $\square$

Because of this result, we can apply the free obstruction theory from the end of Chapter 2. Let us apply this to recover some known results about immersions $S^n \looparrowright \mathbb{R}^{2n}$. To lift such an immersion from $T_1 \text{Emb}(M, N)$ to $T_2 \text{Emb}(M, N)$ is to find an equivariant lift in the diagram

$$\begin{array}{ccc} \partial_2 S^n & \longrightarrow & F_2 \mathbb{R}^{2n} \\ \downarrow & \nearrow & \downarrow \\ F_2 S^n & \longrightarrow & \mathbb{R}^{4n}. \end{array}$$

Since the lower right triangle clearly always commutes up to homotopy, we might as well discard the $\mathbb{R}^{4n}$ and consider the resulting triangle.

We now use our theory to recover the following result, which is due to Smale [Sma59].

Theorem 5.2. *Let $n \geq 3$. Fix an embedding $e : S^n \rightarrow \mathbb{R}^{2n}$ and let $f : S^n \rightarrow \mathbb{R}^{2n}$ be any choice of immersion. The Smale invariant $\Omega(e, f) \in \pi_n V_n \mathbb{R}^{2n}$ is an obstruction to deforming f into an embedding, and it is a complete obstruction. Since the Smale invariant also classifies regular homotopy classes of immersions, this means that f can be deformed into an embedding if and only if it can be deformed into e , i.e. all embeddings are regularly homotopic to e .*

Remark. Smale’s original proof of this result was by showing that the Smale invariant $\Omega(f, f')$ vanishes if and only if f and f' have the same self-intersection number, i.e. the same Wall invariant.

Proof. We study the (equivariant) lifting problem

$$(5.2) \quad \begin{array}{ccc} \partial_2 S^n & \longrightarrow & F_2 \mathbb{R}^{2n} \\ \downarrow & \nearrow & \\ F_2 S^n & & \end{array}$$

where we shall model $\partial_2 S^n$ by STS^n with the induced action from the antipodal action on S^n and $F_2 S^n$ by $DT S^n$ with an analogous action. It is easily seen that the inclusion $STS^n \rightarrow DT S^n$ is a $\mathbb{Z}/2$ -CW-complex; the CW-structure is obtained by filling the spherical fibers over $S^k \subset S^n$ with disks one k at a time. Observe that the bundle $TS^n|_{S^k}$ for $k < n$ is trivial, so we can write the $(n+k)$ -skeleton of this CW-decomposition as $STS^n \cup S^k \times D^n$ for $k < n$.

Notice that $\pi_k F_2 \mathbb{R}^{2n} \cong \pi_k S^{2n-1} = 0$ for $k < 2n-1$. Therefore the obstruction classes h^k that live in $H_{\mathbb{Z}/2}^k(F_2 S^n, \partial_2 S^n; \pi_{k-1} F_2 \mathbb{R}^{2n})$ for $k < 2n$ all vanish. This means we can always pick a lift to the $(2n-1)$ -skeleton, $\sigma : STS^n \cup S^{n-1} \times D^n \rightarrow F_2 \mathbb{R}^{2n}$. Thus, to determine whether a lift in the diagram exists, we need to understand only the obstruction class h^{2n} that lives in $H_{\mathbb{Z}/2}^{2n}(F_2 S^n, \partial_2 S^n; \pi_{2n-1} F_2 \mathbb{R}^{2n})$.

Let us study this obstruction class. It is the obstruction to filling the diagram

$$\begin{array}{ccc} STS^n \cup S^{n-1} \times D^n & \longrightarrow & F_2 \mathbb{R}^{2n} \\ \downarrow & \nearrow & \\ DT S^n & & \end{array}$$

with an equivariant map. A top cell of $DT S^n$ is attached to the $(2n-1)$ -skeleton by gluing onto $STS^n|_{D_n^+} \cup S^{n-1} \times D^n \cong D^n \times S^{n-1} \cup S^{n-1} \times D^n = \partial(D^n \times D^n) \cong S^{2n-1}$. The other cell is obtained from this one by flipping and is glued on via the clutching map of TS^n . So by definition, the obstruction h^{2n} is whether the homotopy class α of σ restricted to this $(2n-1)$ -sphere is zero, i.e. the class $h^{2n} \in H_{\mathbb{Z}/2}^{2n}(F_2 S^n, \partial_2 S^n; \pi_{2n-1} F_2 \mathbb{R}^{2n})$ sends the top equivariant chain to exactly α .

Let us notice that, if we have any two immersions f and $g : S^n \looparrowright \mathbb{R}^{2n}$, we can come up with a regular homotopy to make them agree on one hemisphere, just as in the construction of the Smale invariant. In particular, we can make them agree on the equator $S^{n-1} \subset S^n$. This means that we can assume that the restriction of $STS^n \rightarrow F_2 \mathbb{R}^{2n}$ to $STS^n|_{S^{n-1}}$ does not depend on the choice of immersion, and thus we can also fix a choice of map $S^{n-1} \times D^n \rightarrow F_2 \mathbb{R}^{2n}$ that will define a preferred choice of σ for *any* immersion.

If we start with the embedding e , we know that we can lift the diagram (5.2), by construction, so $\alpha = 0$. Now if we replace e by an immersion f that is not regularly homotopic to e , we replace the map $STS^n|_{D_n^+} \rightarrow F_2 \mathbb{R}^{2n}$ by one such that their difference is nonzero, meaning that if we glue the map $STS^n|_{D_n^+} \rightarrow F_2 \mathbb{R}^{2n}$ coming from e to the one coming from f we get a map $STS^n \rightarrow F_2 \mathbb{R}^{2n}$ that is not null. Since σ will stay the same on $S^{n-1} \times D^n$, in particular we get that α cannot be zero, so h^{2n} does not vanish.

However, we are not done, because we picked a specific lift σ to the $(2n-1)$ -skeleton, while we should prove that the obstruction is nonzero for all choices of lift. Fortunately for us, α , and thus also h^{2n} , depends only on the lift up to non-equivariant homotopy, so we only need to worry about possible non-distinct choices of lift up to usual homotopies. However, such choices are controlled by the ordinary

cohomology groups $H^k(DTS^n, STS^n; \pi_k S^{2n-1})$ for $k \leq 2n - 1$. These are clearly zero for $k < 2n - 1$, and for $k = 2n - 1$ we will get $H^{2n-1}(DTS^n, STS^n; \mathbb{Z})$, which is equal to $H^{n-1}(S^n; \mathbb{Z})$ by the Thom isomorphism for TS^n , and this is clearly zero (as long as $n > 1$). Therefore the obstruction class does not depend on the choice of lift, and we have proven the theorem once we remark that the Smale invariant is exactly the invariant that classifies immersions in the sense of the previous paragraph. $\square$

6. CLOSING REMARKS

In this thesis, we gave an idea of what a theory of deforming immersions into embeddings based on embedding calculus and obstruction theory should look like. We showed that the approach is roughly as follows: first, use the convergence rate of embedding calculus (Theorem 4.1) to determine how many layers you need to lift up, then attack the relevant lifting problems from Theorem 4.5 with obstruction theory. This gives a useful conceptual framework to understand these deformation problems and also allows for computations that do not rely on choosing an immersion with transversal self-intersections. Moreover, it should theoretically be possible to exploit this setup to not just classify embeddings up to regular homotopy, but up to isotopy as well.

To approach such a classification, we use the second statement of equivariant obstruction theory applied to the square (5.1). This will give us (at least modulo some extension problems) a description of the homotopy classes of equivariant lifts. There is a catch, however: such homotopy classes of lifts do not translate directly to isotopy classes of embeddings. Indeed it follows from Theorem 4.5 that the homotopy fiber of $T_n \rightarrow T_{n-1}$ is given by the space of lifts. However, we are not interested in π_0 of the homotopy fiber, we are interested in $\pi_0 T_n$. Therefore, we need to mod out by the boundary map from $\pi_1 T_{n-1}$ to determine the actual isotopy classes of embeddings.

I will mention a few known results about this. For embeddings $S^3 \rightarrow \mathbb{R}^6$, we know (Theorem 5.2) there is only one up to regular homotopy. However, there is $\mathbb{Z}$ -many isotopy classes of such embeddings (in fact, connected sum gives a natural group structure on the isotopy classes where the standard embedding is the identity, and this group is isomorphic to $\mathbb{Z}$). To recover this result from embedding calculus, we would have to go to the third layer, since the map $e_2 : \text{Emb}(S^3, \mathbb{R}^6) \rightarrow T_2 \text{Emb}(S^3, \mathbb{R}^6)$ is only 0-connected.

If we go into the metastable range, we get the following classification result.

Theorem ([HH63], [Sko10]). *Let M be a k -connected oriented m -manifold. For $m \geq 2k + 4$ and $k \geq 0$ we have a one-to-one correspondence*

$$\pi_0 \text{Emb}(M, \mathbb{R}^{2m-k}) \rightarrow H^{m-k-1}(M, \mathbb{Z})$$

when $n - k$ is odd and

$$\pi_0 \text{Emb}(M, \mathbb{R}^{2m-k}) \rightarrow H^{m-k-1}(M, \mathbb{Z}/2)$$

when $n - k$ is even.

In particular, if we take $M = S^m$ and $k = 0$, we obtain that there is a unique isotopy class of embeddings $S^m \rightarrow \mathbb{R}^{2m}$. It should be possible and interesting to recover this and similar classifications from the theory of this thesis.

Let me elaborate on the conceptual aspect a bit further. If one studies the literature on these sorts of problems, one often encounters obstructions that apply step-by-step. For deformation problems, that is, there is some primary obstruction, if that one vanishes there might be some secondary obstruction, if that one vanishes there might be some tertiary obstruction, et cetera. For classification problems, this manifests as the total collection of embeddings fitting in the middle of some short exact sequence, meaning we have to solve an extension problem (or potentially many). The obstruction theory approach illustrates why these phenomena appear: the geometric invariants, at least intuitively, correspond to lifting up certain stages

of the Postnikov tower or lifting up certain layers of embedding calculus. Examples of this already occur when studying embeddings $M^m \rightarrow N^{2m-1}$. For the deformation problem of an immersion $M^m \looparrowright N^{2m-1}$, some step-by-step obstructions are given in [Gra13]. The classification problem for embeddings $M^m \rightarrow \mathbb{R}^{2m-1}$ is considered in [Sko10], and the final result is a kind of exact sequence. It appears that these results could be recontextualized using the approach of this thesis.

Another advantage of this approach is that it also works for a codomain that is not a Euclidean space. In this case we have to deal (on the second layer) with the groups $\pi_n(F_2N \rightarrow N \times N)$ and the $\mathbb{Z}/2$ -action on this. An interesting and to my knowledge not very well-understood problem is to consider the deformation and classification problems for $S^m \rightarrow S^m \times S^m$, for $m \geq 3$. In the case $m = 1$, such embeddings are understood; immersions $S^1 \rightarrow S^1 \times S^1$ can be represented by a pair of integers (p, q) displaying the winding number in both coordinates and an element a of $\mathbb{Z}/2$ that displays whether we have a figure-8 type twist. Such an immersion deforms into an embedding if and only if $a = 0$ and p, q are coprime or both zero, and these embeddings are unique up to isotopy, so we have a classification as well. One can wonder if a similar classification holds for larger m . This is a problem I speculate can be attacked using the approach of this thesis, which should be the topic of further research.

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